

Incendios Forestales

Vicente Bastura | Roberto Carrillo Aguilar | Robert Galpern-Levin | Luis Novoa

UNDERSTANDING PERCEPTIONS OF STAKEHOLDERS IN CUENCA CANTON SURROUNDING WILDFIRES

An Interactive Qualifying Project Report

Submitted to the Faculty of
WORCESTER POLYTECHNIC INSTITUTE
in partial fulfillment of the requirements for the
degree of Bachelor of Science

by
Vicente Bastura
Roberto Carrillo Aguilar
Robert Galpern-Levin
Luis Novoa

Date:
6 March 2026

Project Center:
Cuenca, Ecuador

Project Center Director:
Guilherme Chalhoub Dourado

Sponsor:
Dr. Pablo Geovanny Quichimbo Miguitama
University of Cuenca

Advisors:
Professors Kristin Wobbe, and Edward A. Clancy
Worcester Polytechnic Institute

This proposal represents work of WPI undergraduate students submitted to the faculty as evidence of a degree requirement. WPI routinely publishes these reports on its web site without editorial or peer review. For more information about the projects program at WPI, see http://www.wpi.edu/Academics/Projects.

ABSTRACT

El Cajas National Park in Cuenca, Ecuador plays a vital role in regulating global humidity and providing potable water to nearby cities and communities. In recent years, increasingly frequent and severe wildfires have threatened this park. Sponsored by Dr. Pablo Quichimbo of the University of Cuenca, our goal was to understand community perceptions and knowledge surrounding these wildfires and their effects on soil. We surveyed rural communities and municipal representatives around the Cuenca Canton region. Respondents generally reported limited soil knowledge and held little confidence in fire management policies. We recommend improving education on the effects on soil and the communication of policy.

TABLE OF CONTENTS

ABSTRACT i
TABLE OF CONTENTS ii
LIST OF FIGURES iii
LIST OF TABLES iii
AUTHORSHIP TABLE iv
ACKNOWLEDGEMENTS 1
EXECUTIVE SUMMARY 2
Introduction and Background 2
Methods 2
Results 3
Discussion 5
Conclusions & Recommendations 6
BACKGROUND 7
Drivers and Impacts of Wildfires 7
Limitations of Current Wildfire Management Policies 9
Community Perceptions of Wildfires 12
Effects of Wildfires on Soils 13
METHODS 15
Objective 1: Surveying Rural Populations' Knowledge and Perspectives 15
1.1 Population 15
1.2 Data Collection 15
Objective 2: Surveying Municipal Representatives' Viewpoint 15
2.1 Population 15
2.2 Data Collection 16
Objective 3: Combined Data Analysis for Surveys 17
Objective 4: Integrating Previous Survey Data and Soil Analysis 17
3.1 Population and Data Source 17
3.2 Data Integration and Analysis 17
RESULTS & DISCUSSION 19
Wildfire Knowledge 19
Comparison to Existing Soil Data 23
Perceived Risk 24
Perceptions Surrounding Wildfire Policy 26
CONCLUSIONS 32
Limitations 32
Recommendations 33
BIBLIOGRAPHY 35

APPENDIX A: Rural Community Survey **40**
APPENDIX B: Municipal Representatives Survey **44**
APPENDIX C: UConCiencia Unformatted Post **48**

LIST OF FIGURES

Figure 1: *Weighted Rankings of Responsibility for Wildfire Prevention* 5
Figure 2: *Integrated Fire Management Roadmap* 10
Figure 3: *Map of All Sites for Data Collection* 16
Figure 4: *Perceived Wildfire Information Familiarity* 20
Figure 5: *Perceived Knowledge of Wildfire Effects on Soil* 21
Figure 6: *Perceived Primary Cause of Wildfires* 21
Figure 7: *Perceived Changes in Wildfire Frequency in Recent Years* 25
Figure 8: *Perceived Severity of Wildfire's Effects on Nearby Communities* 26
Figure 9: *Perceived Level of Knowledge on Wildfire Policy* 27
Figure 10: *Perceived Policy Effectiveness* 28
Figure 11: *Perceived Level of Wildfires Preventability* 28
Figure 12: *Perceived Difficulty of Policy Creation* 29
Figure 13: *Perceived Influence of Soil Effects from Wildfires on Policy* 30

LIST OF TABLES

Table 1: *Statistical Analysis of Common Multiple Choice Questions* 3
Table 2: *Reported General Effects of Wildfires* 4
Table 3: *Reported Effects of Wildfires on Soil* 4
Table 4: *The Economic Toll of Fires in Brazil* 8

AUTHORSHIP TABLE

Section	Primary Author	Editor 1	Editor 2	Editor 3
I. Cover Page	LN	RGL	RCA	VB
II. Abstract	RCA	RGL	VB	LN
III. Acknowledgements	LN	RGL	RCA	VB
IV. Executive Summary	VB	RCA	RGL	LN
1.0 Background				
1.1 Drivers and Impacts of Wildfires	RGL	RCA	VB	LN
1.2 Limitations of Current Wildfire Management Policies	RCA	LN	RGL	VB
1.3 Community Perceptions of Wildfires	VB	RCA	RGL	LN
1.4 Effects of Wildfires on Soils	LN	RGL	VB	RCA
2.0 Methods				
2.1 Objective 1: Surveying Rural Populations' Knowledge and Perspectives	RCA	RGL	LN	VB
2.2 Objective 2: Surveying Municipal Representatives' Viewpoint	RGL	VB	LN	RCA
2.3 Objective 3: Combined Data	RGL	VB	RCA	LN

Analysis for Surveys				
2.4 Objective 4: Integrating Previous Survey Data and Soil Analysis	RGL	VB	RCA	LN
3.0 Results & Discussion				
3.1 Intro	VB	RGL	RCA	LN
3.2 Wildfire Knowledge	VB	RGL	RCA	LN
3.3 Comparison to Existing Soil Data	VB	RCA	RGL	LN
3.4 Perceived Risk	RCA	RGL	VB	LN
3.5 Perceptions Surrounding Wildfire Policy	RGL	RCA	VB	LN
4.0 Conclusion	RCA	RGL	VB	LN
4.1 Limitations	RGL	RCA	VB	LN
4.2 Recommendations	RGL	RCA	VB	LN
5.0 Appendices				
5.1 Appendix A: Rural Community Survey	VB	RGL	RCA	LN
5.2 Appendix B: Municipal Representatives Survey	RCA	RGL	VB	LN
5.3 Appendix C: UCon Ciencia Unformatted Post	RCA	RGL	VB	LN
Analysis	VB	RCA	LN	RGL
Making Figures and Tables	VB	RCA	RGL	LN

ACKNOWLEDGEMENTS

We would like to sincerely thank Dr. Pablo Geovanny Quichimbo Miguitama for his guidance, resources, and collaboration with the project, as well as the University of Cuenca for giving us the opportunity to conduct this research. Noelia Pañi, Jilson Ulloa, Angelita Quinde, and Tania Jérez are students at the University of Cuenca who worked with us throughout the project and were a great help. We are thankful to WPI for giving us the opportunity to travel, study abroad, and contribute to a meaningful project. Our professors from WPI, Kris Wobbe and Ted Clancy, were crucial to this project, giving us consistent feedback and helping us grow as students and writers. Cuenca program site director Guilherme Dourado was an immense resource for this project, helping us develop our project and work with our sponsor. Finally, we would like to thank all of the community members who took our surveys, giving us their valuable time and knowledge.

EXECUTIVE SUMMARY

Introduction and Background

El Cajas National Park in Cuenca, Ecuador, is vital to the environment and its surrounding communities, helping to regulate global humidity, and importantly for locals, it captures rainwater and provides potable water. Unfortunately, El Cajas has experienced increasingly frequent and severe wildfires in recent years, threatening its ecosystems and the critical water resources it provides to these communities and South America as a whole. Wildfires significantly affect the soil, reducing ecosystem recovery, threatening water quality for downstream communities, and inhibiting the soil's ability to retain significant moisture. Many community members have been affected and have valuable perspectives on these fires.

Understanding community perceptions of wildfires, such as perceived risks and knowledge, is important to provide a basis for comparison with known effects of wildfires. Differences between perceived and actual impacts identify important matters that need to be addressed through more awareness and policy. Perceptions influence how people act and need to be aligned with reality, or they lead to ineffective actions for the prevention and management of wildfires. A better understanding of science-driven policy may also improve a community's trust in municipal organizations and policy. Additionally, differences between community and technical knowledge provide insight into creating better community resources. (Calvani et al., 2025).

Methods

We developed four objectives to obtain perceptions and compare them to the technical analysis of wildfire effects.

Objective 1: Surveying rural populations' knowledge and perspectives on wildfires
Objective 2: Surveying municipal representatives' viewpoints on wildfires
Objective 3: Combined data analysis for surveys
Objective 4: Integrating previous survey data and soil analysis on wildfires

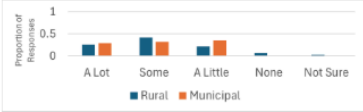
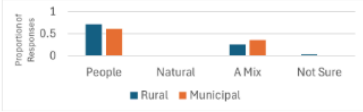
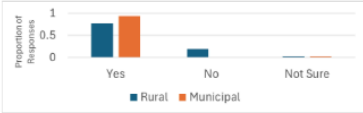
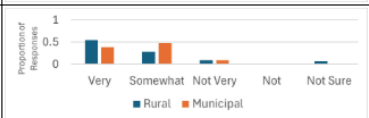
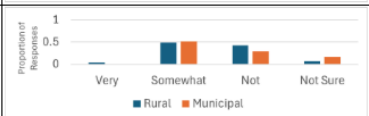
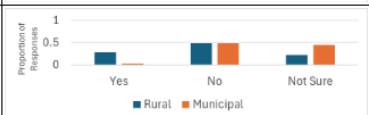
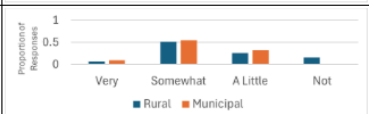
We used convenience sampling due to our seven-week time constraint, as limited resources rendered other sampling options infeasible. Surveys with overlapping questions were designed for each of the two populations: rural communities and municipal representatives. These surveys had questions aimed at gauging perceived risk, perception of policy, level of general wildfire knowledge, and some specific soil knowledge. The municipal survey additionally had questions aimed at the subject's understanding of policy creation. A full list of survey questions can be found in Appendices A and B. For all surveys, written consent was obtained from participants beforehand. To administer the rural population survey, we travelled to different open-air markets within Cuenca. To administer the municipal survey, we travelled to the offices of different municipal organizations. Objective 3 compared all common questions using statistical analysis methods. For Objective 4, lab results from previous studies on the effects of wildfires on soil were used to compare with the qualitative data. Completing these objectives allowed us to discover gaps in knowledge, perceptions of policy, and comparisons of perceived and known impacts. This key information guides recommendations for future policy, community resources, and research.

Results

In total, we surveyed 31 members of rural communities and 30 municipal representatives. The results highlight key patterns related to knowledge levels, perceived risks, and perceptions of policy, as well as comparing rural and municipal populations. Initially, we statistically compared responses from the eight closed-ended and two open-ended questions that were in the surveys for both rural and municipal populations. Seven of the closed-ended questions were multiple choice with varying response options. There was one additional closed-ended ranked question that was converted into a (weighted) Borda count. The two open-ended responses were converted to a frequency table. Using the chi-squared test individually for each question, we found that four out of the 10 questions had statistically significant differences between groups (using a per-test Type I error rate of 0.05). However, after accounting for the family-wise error using the Holm-Bonferroni adjustment for all 10 questions, we found that there were no questions with a statistically significant difference in responses (using a family-wise Type I error rate of 0.05). Response bar graphs for all common closed-ended questions, along with the p-values and adjusted significance thresholds for these questions, are listed in Table 1. The two open-ended question frequency tables are shown in Tables 2 and 3. These results are discussed below.

Table 1

Statistical Analysis of Common Multiple Choice Questions

Question	Graph	P-value & Threshold
How much do you know about wildfires?		$P = 0.3640$ $\alpha_i = 0.0250$
What is the main cause of wildfires?		$P = 0.1953$ $\alpha_i = 0.0100$
Have wildfires become more frequent?		$P = 0.0396$ $\alpha_i = 0.0071$
How preventable are wildfires?		$P = 0.2264$ $\alpha_i = 0.0125$
How knowledgeable are you on wildfire policy?		$P = 0.3871$ $\alpha_i = 0.0500$
Is wildfire policy effective?		$P = 0.0128$ $\alpha_i = 0.0050$
How knowledgeable are you on wildfires' effect on soil?		$P = 0.1294$ $\alpha_i = 0.0083$

Note. This figure shows all seven multiple choice questions and the results of the chi-squared analysis with the Holm-Bonferroni adjustment for all 10 questions that were common to the surveys of rural and municipal populations. P is the p-value, and α_i is the adjusted threshold for each question using a family-wise Type I error rate of 0.05.

We used grounded theory to determine categories for the answers to the open-ended question about the general effects of wildfires. Subsequently, frequency analysis was used to convert this qualitative data into a quantitative frequency table for both populations, where the chi-squared test was applied. Similar categories were combined so that there were common categories between the groups, such as climate impacts and environment, to improve the quality of analysis. Similar categories allowed the tables for each population to be combined into one. The final table for the general effects of wildfires can be seen in Table 2. A chi-squared test gave a p-value of 0.0333, which is greater than the adjusted significance threshold of 0.0063.

Table 2

Reported General Effects of Wildfires

Reported General Effects of Wildfires		
Category	**Rural Frequency**	**Municipal Frequency**
Animals	15	14
Water	12	5
Vegetation	10	15
Air quality / Environment / Climate Impacts	8	11
Biodiversity / Ecosystem	5	12
Agriculture / Livelihoods	4	0
Soil degradation	3	9

Note. This table shows the frequency of categories mentioned within the open-ended question pertaining to the general effects of wildfires for rural and municipal populations.

Similarly, for the other open-ended question on soil-specific effects of wildfires, frequency analysis was used, similar categories were combined, and the tables were grouped into one with both populations. The combined table can be seen in Table 3. A chi-squared test gave a p-value of 0.2974, which is greater than the adjusted significance threshold of 0.0167.

Table 3

Reported Effects of Wildfires on Soil

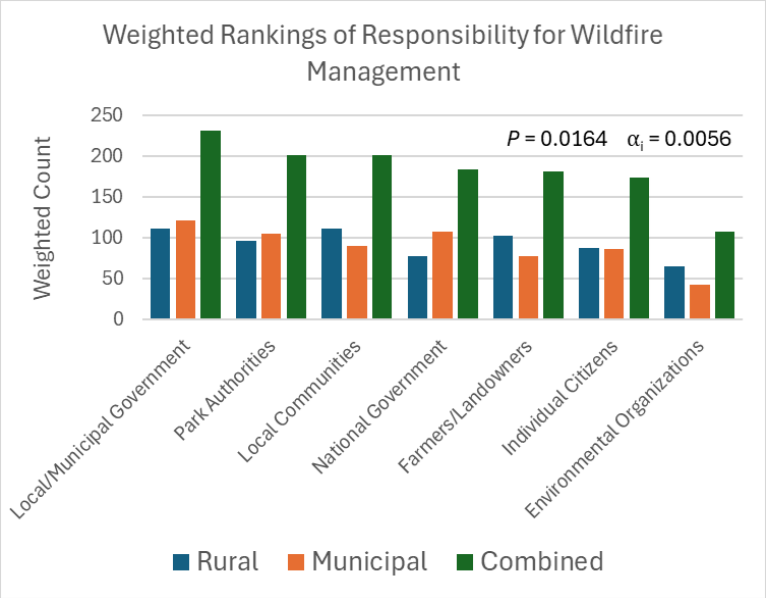
Reported Effects of Wildfires on Soil		
Category	**Rural Frequency**	**Municipal Frequency**
Vegetation recovery	8	9
Soil fertility / Nutrients	7	12
Soil erosion / Physical degradation	6	13
Soil microorganisms / Biology	3	11
Water retention	3	10
Biodiversity / Ecosystem	0	8
Soil structure and chemical properties	5	8

Note. This table shows the frequency of categories mentioned within the open-ended question pertaining to the specific effects of wildfires on soil for rural and municipal populations.

Individuals were asked to rank seven options from most to least responsible for wildfire management. For analysis, a Borda count was used to produce a weighted count, where a higher responsibility ranking is associated with a higher count. Figure 1 shows the total count for the rural and municipal populations, as well as the combined count.

Figure 1

Weighted Rankings of Responsibility for Wildfire Prevention



Note. This chart shows the weighted results of what rural and municipal populations believe is most responsible for managing wildfires, presented in descending combined weighted count. The X-axis is the response options for possible responsible parties, and the Y-axis is the weighted count of the survey results.

Discussion

Several key findings and insights emerged as a result of the surveys and analysis, from both the questions that were common between the rural population and the municipal representatives, and those questions that were distinct. For knowledge, both groups generally believe themselves to be at least somewhat knowledgeable about soil-specific effects from wildfires. Only 16% of rural respondents and no municipal representatives reported not being knowledgeable. Additionally, most respondents knew that the primary cause of wildfires was humans, consistent with a case study that found that humans caused 87% of wildfire ignitions (Harrison et al., 2025). To further improve our understanding of community knowledge, the open-ended questions gave deeper insight into the effects of wildfires on soils. When considering specific changes observed in soils post-wildfire (Table 2), rural populations noted the effects on vegetation recovery, soil nutrients, and soil erosion as their main concerns. Municipal representatives, on the other hand, stressed the physical degradation of soil the most, followed by impacts on soil fertility and microorganisms.

The reported effects from Table 2 were compared with quantitative data from a 2022 study on the effects of wildfire severity on the physical-chemical properties of soil in southern Ecuador, which stated that high-severity burns in 2019 were "negatively correlated with variables related to soil chemical fertility" (Carrión-Paladines et al., 2022). Consequently, we found that people's perceptions with respect to soil degradation align with reality. Significant parallels can be drawn between our findings and previous studies. Agricultural producers generally rely more heavily on qualitative indicators when making decisions based on soil. For instance, Amazonians may determine that certain black or brown soil contains earthworms based on the color (Jiménez et al., 2024). From survey responses, we likewise found that farmers identify soil degradation based on visually observable soil characteristics.

This is further corroborated by our respondents' answers congregating on concerns of soil losing its fertility, which concurs with the finding of decreased soil organic carbon and pH levels in burn areas (Cáceres, 2022). Survey respondents also noted a loss of plant life and an affected recovery rate following a fire. Cáceres provides a quantitative foundation for this observation by mentioning that burn sites typically experience 8.33 times less aerial plant biomass when compared with unaffected areas for a period of 18 months after a fire (Cáceres, 2022).

When considering perceived risk, the survey respondents claimed that the frequency of wildfires had increased in recent years, with over 75% agreement between both groups. A study published in 2024 found that wildfire frequency over agricultural land had followed an upwards trend between 2001 and 2020, totaling 440, 772 ha over the full period (Vaca et al., 2024), supporting this idea. Of those surveyed, 84% considered the risk of wildfires to be serious or very serious, with only 6% responding that a small or nonexistent level of risk exists.

Pertaining to policy, respondents had a lack of deep knowledge of fire management as well as an inherent distrust in the effectiveness of wildfire policy. This could be due to numerous factors, including issues with the current political landscape and distrust of the government as a whole. However, respondents also said that they believed wildfires are preventable, highlighting that the issue is not a lack of belief in a potential solution, but a lack of belief in the current solution itself. Making the process for creating policy easier and more transparent, as well as increasing widespread understanding of said policy, would increase support of wildfire management policies. Lastly, Figure 1 shows that although there are some differences in ranked responsibility, it is largely a shared responsibility between all groups to determine solutions.

Conclusions & Recommendations

Our main goal for this project was to better comprehend the perceptions and experiences relative to wildfires of our target populations in Cuenca Canton. Using surveys, we collected data from key stakeholders: members of rural communities around Cuenca and municipal representatives. Once the data was collected, we completed a thorough statistical analysis comparing and contrasting the responses of rural community members and municipal representatives. In addition, we used existing laboratory soil data from previous studies as well as previous surveying efforts to improve the robustness of our findings.

Our data suggested that gaps exist among both rural community members' and municipal representatives' knowledge on the technical impacts on soil post fire. Stakeholder groups also hold low confidence in fire management policies and their effectiveness. By contrasting our findings with previous research conducted in the same region, these studies together underscore the importance of strengthening connections between scientific research, community knowledge, and municipal decision-making. Our findings provide a baseline for future iterations and extensions of this study.

Based on our findings, we developed the following recommendations for how our results can help the community.

1. More education around wildfires and their technical effects on soil is needed.
2. Policy should integrate community knowledge and scientific findings.
3. Wildfire prevention policy should be more widely understood and more clearly communicated to the public.
4. Our study could help guide further research into similar studies with larger statistical power.

BACKGROUND

Drivers and Impacts of Wildfires

Over the past few decades, wildfires have grown both more frequent and more intense, resulting in significant environmental damage, economic losses, and public health impacts across many regions (Eufemia et al., 2022). These wildfires are causing increasing environmental damage, economic loss, and public health challenges across many regions. This trend has been prominent in South America, where the climate phenomena and changes in temperature intensify wildfire risk.

Large regions of South America experience normal wet and dry seasons throughout the year. This weather pattern is disrupted by the "El Niño-Southern Oscillation", where the change in temperature of waters in the Pacific Ocean disrupts the atmosphere and changes rain patterns across South America, leading to floods and droughts (Reis et al., 2025). The El Niño-Southern Oscillation has existed for thousands of years, but as climate change increases due to human activity, the effects also increase (Reis et al., 2025). Dry season lengths are extended, and droughts become more severe. As a result, wildfires in these regions ignite more easily and spread rapidly during these seasons. These fires become harder to contain and more destructive.

In addition to the climate, changes to ecosystems across South America have created an environment where fires can spread rapidly and are difficult to fight. Forest fragmentation is a process in which large forests are broken up into smaller, more isolated areas. Deforestation, infrastructure development, and the increase in land being used for agriculture are all causes of forest fragmentation (Noble, Gilroy, & Peres, 2025). The Andean highlands region has fragmented ecosystems that dry out faster during droughts, increasing the risk of wildfires (Armenteras et al., 2017). In addition, these environments are close to human populated areas allowing for small human-ignited fires to quickly grow out of control (Armenteras et al., 2017).

Wildfires affect the environment in ways that impact nearby communities long after the burn. When forests are destroyed, the area becomes more exposed, leading to higher surface temperatures and greater temperature extremes between day and night (Carrión-Paladines et al., 2023). These changes disrupt wildlife habitats, reducing food resources and shelter options, often forcing animals to migrate. This, in turn, affects the nearby communities that rely on these

animals. In addition, the environmental destruction also leaves areas more vulnerable. During heavy rain, burned slopes are more likely to experience landslides, putting homes, roads, and potable water sources at risk (Qin et al., 2024).

As wildfires rip through vegetation and forests with increasing intensity and scale, the damage and economic costs continue to rise (Sobreira et al., 2025). In Table 4 below, the cost of repairing the Brazilian ecosystem and biodiversity damaged by wildfires from January to August 2024 was over 22 billion USD. This increased economic strain caused by wildfires compounds over time and slows recovery efforts.

Table 4

The Economic Toll of Fires in Brazil

Major question	Impact	Costs	Estimates descript
What is the cost of wildfires to public health?	Healthcare	US$ 10.2 million (cardiovascular issues) US$ 4.4 million (respiratory issues)	Annual estimation of hospitalizations, with a 21% increase in cardiovascular and 23% in respiratory issues due to wildfire smoke, using 2000–2023 averages (2,046,234 hospitalizations) at $63.25 per day.
What is the cost of wildfires to small-scale family farming?	Small-scale family farming	US$ 75 million	Estimates based on data provided by the National Institute for Colonization and Agrarian Reform (INCRA)
What is the cost of wildfires to energy infrastructure?	Energy infrastructure	US$130 million	The costs involved include not only the repair of direct damage but also economic losses due to service disruptions and emergency response efforts.
What is the cost of wildfires supply chains?	Supply chains	No reliable estimate.	–
What is the cost of wildfires to transport?	Transport	No reliable estimate.	–
What would be the estimated costs for the impacts of wildfires on biodiversity and ecosystem services?	Biodiversity	US$ 14.7 billion	Based on an approach that ponders that a half of the global GDP directly or indirectly depends on nature and ecosystem services, considering the framework available in The World Economic Forum's Biodiversity Finance Initiative (see https://www.weforum.org/ ↗).
What would be the costs for restoration considering this burned area?	Ecosystem restoration	From US$ 7 to 8 billion	Based on studies concerning the value of ecological restoration practices in Brazilian ecosystems (e.g., Brancalion et al., 2019).

Note. Economic costs of numerous different impacts caused by wildfires in Brazil, as well as the description of the costs (Sobreira et al., 2025).

Air pollution caused by smoke from these fires has become a major public health concern across South America, as wildfire smoke contains high concentrations of PM2.5, small air pollutant particles, and carbon monoxide that cause widespread respiratory impacts even far from the initial fires. Exposure has been shown to cause asthma attacks, affect lung capacity (Urrutia-Pereira et al., 2021), and distinctly affect vulnerable groups such as children, as seen in a study from Quito, Ecuador (López-Gil, 2025). In the Andean region, where droughts are lasting longer and becoming more intense, communities are exposed to wildfire smoke for extended periods, contributing to a growing number of South Americans developing respiratory issues. A study of over two million hospital admissions between 2008 and 2018 found that wildfire-related air pollution has led to a 20% increase in respiratory problems (Requia et al., 2021), causing more hospital visits and putting additional strain on health systems that are already frequently overburdened.

Limitations of Current Wildfire Management Policies

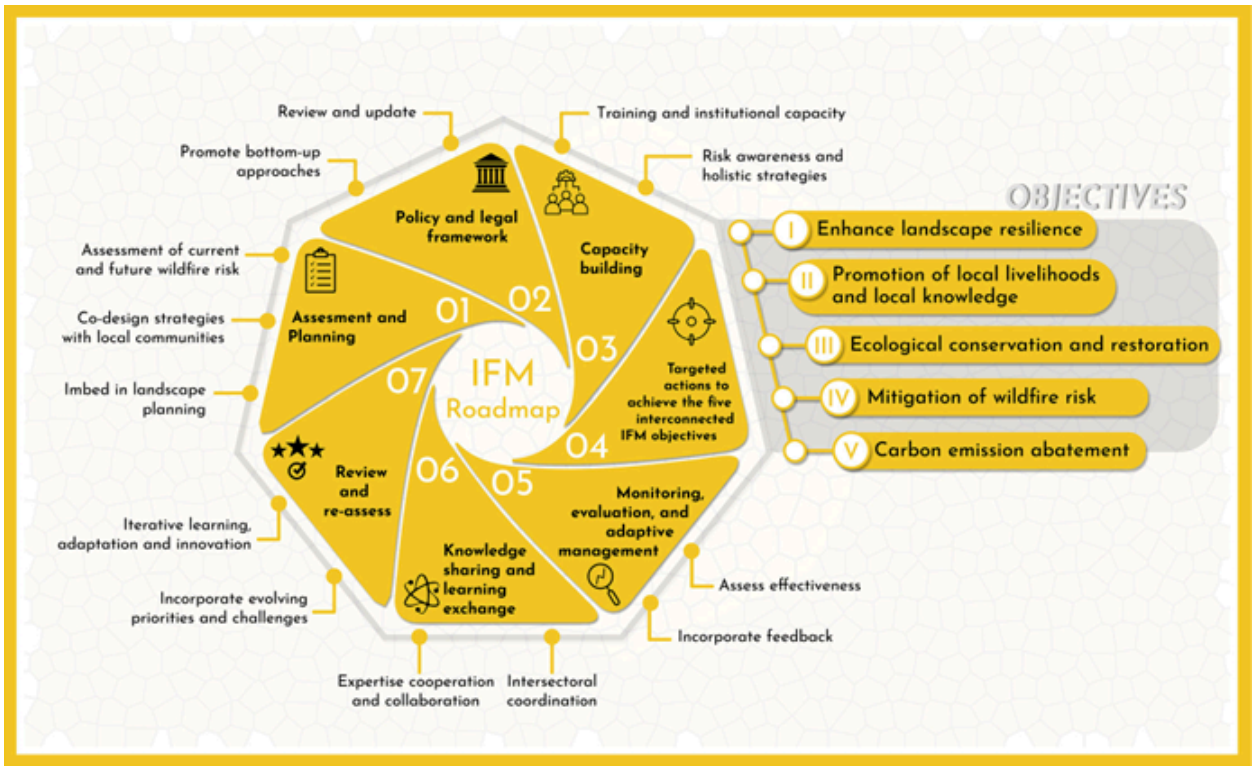
Understanding the effects of wildfires is essential to creating effective fire management policies and strategies. As climate change's amplifying effect on wildfires continues, so must the development and implementation of these policies.

While there are fire management policies in place in South America to mitigate and manage fires, a number of unaddressed or poorly addressed issues remain. These issues are evident in multiple aspects of policy, such as the lack of concrete standards across regions, conflict with indigenous fire practices, financial and operations gaps, and the disregard for fire severity in policy. More comprehensive and streamlined fire management approaches are recommended than are currently present (Oliveras Menor et al., 2025). A main issue with these policies is their tendency towards suppressing wildfires once active rather than incorporating measures for their prevention (Mistry et al., 2016). Proposed preventive strategies include maintaining the amount of carbon stored in vegetation, identifying high-risk areas more susceptible to burning, and managing vegetation for a lower risk of wildfires (Moura et al., 2025).

Unfortunately, fire policies at present are applied very differently across regions and do not all adequately meet regional standards. As an example, the sizable differences in the ACN (Andean Community of Nations) countries' individual policies across categories such as prevention, indigenous fire use, and post-fire restoration, among others, are not in line with the body's regional initiatives. Bolivia, Colombia, and Venezuela, for example, have already proposed bills to implement national fire management policies, but these are still awaiting approval. Mexico and Brazil, on the other hand, have both already approved federal laws on forms of integrated fire management, and Mexico has a national plan that recognizes both the ecological and social role of fires. A proposed solution to solidify standards for fire management across countries in South America is the Andean Directive for Integrated Fire Management (ADIFM), which would effectively produce a more optimal system for managing fire if implemented correctly (Oliveras Menor et al., 2025).

Figure 2

Integrated Fire Management Roadmap



Note. The Andean Community of Nations' roadmap that cycles through the seven core steps and objectives to adapt to and mitigate altered fire regimes through the use of IFM (Integrated Fire Management) strategies (Oliveras Menor et al., 2025).

Independent of inconsistencies in policy across regions, future frameworks should additionally account for traditional indigenous fire practices, such as controlled cultural burns used to regenerate pasture, stimulate new plant growth, and maintain paramo ecosystems in the Andes region. This clash, whether it be with disjointed or standardized fire management law, is worth noting because traditional methods often end up criminalized due to misunderstanding. In many cases, traditional fire use receives no recognition at the time of creating policy, resulting in top-down policies that range from ineffective implementation to rendering indigenous fire management and use illegal in some instances (Mistry et al., 2016). In the future, the tradition in question should be written into policy via co-creation with local communities. One possible mechanism would be for indigenous communities to attend participant platforms to help develop action plans for fire management (Correa-Quezada et al., 2025).

In addition to regional inconsistency and disregard for indigenous knowledge, fire policies face challenges in the form of both financial and operational gaps. On the financial front, the systems put in place to proactively prevent wildfires often do not have stable sources of funding at their disposal. This is the case in many South American countries, where “specific funding” surrounding fire management policy is rated as weak (Oliveras Menor et al., 2025). Policies may be legally effective, but quite inadequately and ineffectively enforced in actual practice. This weak and negligent enforcement only serves to further the consequences of uncontrolled fires. One such example is failure to follow through with enforcing the payment of debts and fines due to pressure from the industries on the paying end (Mistry et al., 2016).

A final issue to consider is that the factor of fire severity should play a role in the evaluation of fire-centric ecological impacts. That is, many fire management policies tend towards the “zero-fire” approach, wherein any degree of fire is suppressed in a black-and-white manner regardless of severity or intent (Oliveras Menor et al., 2025). In actual fact, low-severity, controlled fires can be beneficial for the environment. To better illustrate the point with an example, patch mosaic burning is sometimes employed where the savanna biome transitions to the forest biome so as to shield the forest from possible large-scale wildfires owing to the long-term buildup of fuels (Eufemia et al., 2022).

Further, the severity of fire should be taken into account for the development of fire management strategies and any related sanctions. Wildfires can be categorized as either

low-severity or high-severity fires (Oliveras Menor et al., 2025). Typically, these low-severity fires are either controlled by intentional human action or by a part of the natural dynamics of ecosystems. Due to this fact, much plant life has adapted to low-severity wildfires by developing fire-induced mechanisms (Carrión-Paladines, 2022). Contained within the preventive measures facet of IFM (Integrated Fire Management), prescribed burns using low-severity fires, more common than naturally occurring low-severity fires, can have the effect of enriching the soil with nutrients (Oliveras Menor et al., 2025). These controlled burns are usually applied at spaced time intervals to allow the ecosystem sufficient time to recover. To this end, fire severity can form part of the base upon which adaptive management strategies function. Along with other factors, such as the evolving climate, considering fire severity leads to better-managed vegetation and significantly reduced risk of high-intensity burns such as wildfires (Moura et al., 2025).

Community Perceptions of Wildfires

Understanding community perceptions of wildfires, such as perceived risks and knowledge, is important to provide a basis for comparison with known effects of wildfires. Differences between perceived and actual risk identify important matters that need to be addressed through more awareness and policy. Perceptions influence how people act and must be aligned with reality; otherwise, they can lead to ineffective prevention and management efforts. A better understanding of science-driven policy may also improve a community's trust in municipal organizations and fire management. Additionally, differences between community and technical knowledge provide insight into creating better community resources. (Calvani et al., 2025)

Perceived risk is how individuals within a community anticipate how wildfires will personally affect them. However, this may vary from the actual risk, which is identified through scientific studies and data. When perceptions lack alignment with reality within local communities, it can lead to underpreparedness and excess fear. When perceptions are unaligned with reality within municipal organizations, it can lead to less effective policy and poor resource distribution towards key issues. Understanding actual risks for comparison with the community's perception allows these problems to be addressed. (Calvani et al., 2025)

As described above, wildfires pose a serious health and economic risk to surrounding communities. Health risks may include increased asthma incidents and decreased lung capacity. Additionally, economic damage may be done to homes and infrastructure. Understanding these

risks not only allows individuals to make educated decisions to protect their health and economic stability but can also support effective policy making. Proper risk assessment prompts policy that can most effectively mitigate these effects. An example of this is a 2011 study, which used an assessment of wildfire risk to prioritize mitigation measures and enable risk-supported policy creation (Thompson et al., 2011).

While perceived risk reflects how communities evaluate wildfire threats, assessment of community knowledge reveals gaps in knowledge that influence engagement with mitigation strategies. Enhancing what the community knows about wildfire prevention and management increases community participation. Increased community participation is key to maintaining wildfire management practices effectively long term (Santiago-Gómez et al., 2023). Alongside this, the knowledge of policymakers can contribute to the impact they make. Perceptions influence what policy is targeting, but knowledge influences the specific effectiveness. Both perceptions and knowledge are important factors in creating an overall effective policy.

Effects of Wildfires on Soils

Unlike low-severity fires, which favor the regeneration of grass and plants and bolster the nutrients of soil, high-severity fires are far more dangerous and can cause far more damage (Carrión-Paladines, 2022). Due to their heightened severity, they are much more difficult to control, with an increased risk of spreading and increased soil impact over a longer period. Severe wildfires cause far more changes to the properties of soils in ecosystems, such as the removal of organic matter, deterioration of soil structure, and a considerable loss of nutrients (Certini, 2005). While 85% or more organic matter remains in soil after low-severity fires, less than 20% is left after a high-severity fire, altering the soil to a point of unusability (Nelson et al., 2024).

One major effect high-severity wildfires have is reducing its ability to properly absorb water, making it hydrophobic, meaning it is unable to properly absorb water. This change is highly significant since when soil is hydrophobic, rain causes erosion and runoff rather than being absorbed by the soil. Hydrophobic soil is especially problematic in El Cajas National Park in Ecuador. The paramo ecosystem in the park has soil that functions as a sponge, absorbing water from rainfall and the air. This process is crucial to regulating the humidity of the region and maintaining proper weather patterns (Berrones Guapulema et al., 2022). This is especially

important to the nearby city of Cuenca, which is one of the few cities in South America that has drinkable tap water due to the potable water provided by El Cajas (Arcentales-Ríos et al., 2022). Runoff due to affected soil threatens this drinking supply.

While prescribed fires can help improve agriculture and forest sustainability, high-severity fires are the complete opposite, destroying the possibility of agriculture. As stated previously, wildfires can cause a plethora of effects on communities, such as pollution and health concerns. It is also important to acknowledge that agriculture is critically important to the livelihood of many communities and families. Even without the health or pollution concerns, the plot of land that is burned will be left unusable for several years, resulting in long-lasting damage.

METHODS

Objective 1: Surveying Rural Populations' Knowledge and Perspectives

1.1 Population

We surveyed subjects from rural communities living in Cuenca Canton, Ecuador. The population consisted of market vendors, who are directly involved in agricultural production. Our goal in surveying these agricultural producers was to understand their perceptions and knowledge of wildfires and policy, as well as the specific effects of wildfires and changes in the soil they have witnessed.

1.2 Data Collection

Paired with volunteer students from the University of Cuenca, we conducted a convenience sample of vendors at different open-air markets around Cuenca using both digital and physical formats. The survey (full list of questions in Appendix A) contained a mix of open and closed-ended questions focused on perceived risk, views on policy, if and how participants were personally affected by fires, and understanding their perceptions and knowledge surrounding wildfires. Additionally, there was a smaller focus on their soil-specific knowledge about wildfires. We worked with our sponsor to ensure all questions were correctly translated into Spanish and culturally appropriate. We made online surveys using KoboToolbox, then printed paper surveys to have both options available. All participants first provided written informed consent (WPI IRB approval 26-0380). The surveys took 5-10 minutes to complete, with the entire interaction lasting about 15 minutes, including the informed consent and an introduction.

Objective 2: Surveying Municipal Representatives' Viewpoint

2.1 Population

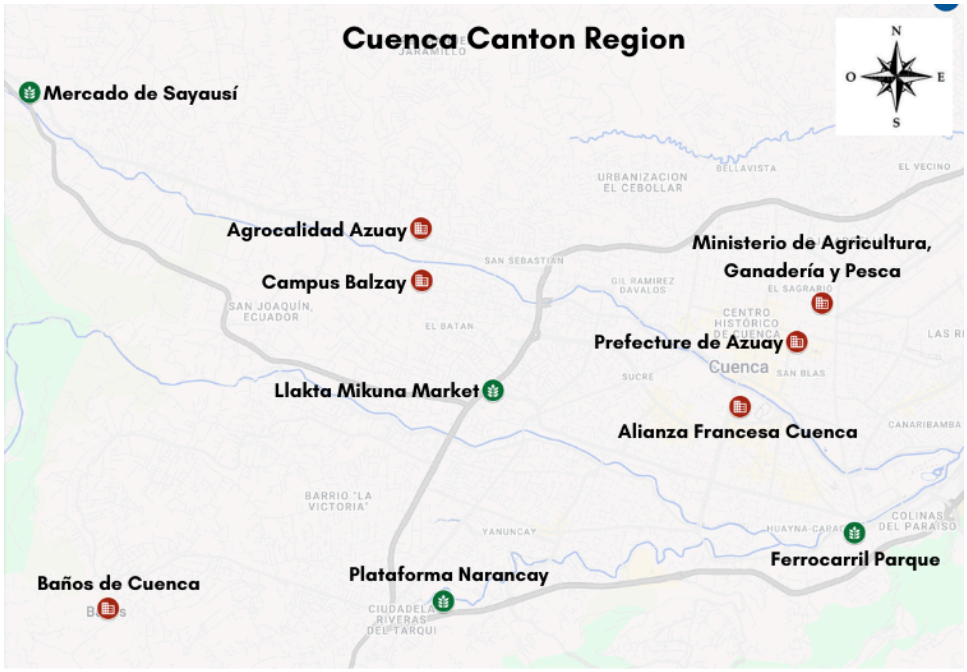
Municipal representatives were surveyed to gain a better understanding of the institutional perspective on fire management and its impacts on soil. These representatives provided a different, more governmental point of view than other stakeholders. Through our survey, we wanted to gain a better understanding of their perceptions and knowledge of wildfires and policy, as well as the specific effects of wildfires and changes in the soil they have witnessed.

2.2 Data Collection

Our method of data collection for municipal representatives was similar to that used for rural communities, but it included a few notable changes. We used surveys that contained 10 questions in common with the survey for rural communities, with additional questions specific to municipal representatives' role, influence in policy, and the difficulties that come with creating new policies (full list of questions in Appendix B). We set up appointments with offices around Cuenca through our sponsor's connections. We administered the surveys in person with students from the University of Cuenca. All participants first provided written informed consent (WPI IRB approval 26-0380). The surveys took 5-10 minutes to complete, with the entire interaction lasting about 15 minutes, including the informed consent and an introduction.

Figure 3

Map of All Sites for Data Collection



Note. This map contains all sites around the Cuenca Canton region that we visited to collect our data. Green icons are markets where rural surveys were administered, and red icons are municipal buildings where municipal surveys were administered.

Objective 3: Combined Data Analysis for Surveys

We logged all surveys that were done on paper into KoboToolbox to have all the data in one place. For closed-ended questions, we used descriptive statistical analysis to determine the distribution of responses for each question in both municipal and rural surveys. For open-ended questions, we utilized grounded theory to extract key terms to find recurring themes and identify consistencies between the stories of individuals. Subsequently, frequency analysis was used every time each category was mentioned within a response, therefore converting this qualitative data into a quantitative frequency table, to which the chi-squared test was applied. To determine if groups had differing responses, we used chi-squared tests to determine statistical significance. Family-wise error was accounted for with the Holm-Bonferroni adjustment, which was applied to all common questions. This accumulated into triangulation to identify recurring themes across both groups.

Objective 4: Integrating Previous Survey Data and Soil Analysis

3.1 Population and Data Source

For this objective, we utilized two previous reports conducted by our sponsor. The first was a 2022 thesis by Juan Carlos Cáceres, who analyzed soil samples collected in June 2021 from a site burned in October 2019 (Cáceres, 2022). The second was a journal article written by our sponsor in conjunction with other academics in 2024 (Jiménez et al., 2024). This article surveyed farmers to understand their knowledge of the impacts of wildfires on soil in the Ecuadorian Amazon.

3.2 Data Integration and Analysis

We conducted a structured comparison between our survey findings and the 2024 article data (Jiménez et al., 2024). We reviewed the qualitative survey responses from the earlier study to identify recurring themes and contrasting results with our data. We also incorporated our data from municipal representatives to attain new findings not possible with only farmer survey responses from the prior study (Jiménez et al., 2024).

In addition, we examined the laboratory soil results from Cáceres thesis (Cáceres, 2022), focusing on physical and chemical changes following wildfires. We contrasted these findings with the soil impacts most commonly identified by participants in our study. This process

allowed us to identify consistencies between scientific measurements and stakeholder perceptions, as well as gaps in knowledge where technical changes in the soil were not widely recognized.

RESULTS & DISCUSSION

This study was aimed at understanding the perceptions and knowledge of key community members, including rural populations and municipal representatives, regarding wildfires in and around El Cajas National Park in Cuenca, Ecuador. Understanding perceptions of community stakeholders is crucial as it allows us to see gaps in knowledge and places where work is needed. We also focused on understanding current perceptions around wildfire prevention policy. We accomplished these goals by administering surveys to population members and contrasted our findings with survey and soil analysis data from previous academic papers from our sponsor and his students. This project's main goals are to inform future research efforts at the University of Cuenca, find gaps in knowledge where more education could be beneficial, and understand where more transparency is needed in the wildfire prevention process.

In total, we surveyed 31 members of rural communities and 30 municipal representatives. The results highlight key patterns related to knowledge levels, perceived risks, and perceptions of policy, as well as differences between rural and municipal populations. We statistically compared responses from the eight closed-ended and two open-ended questions that were the same between rural and municipal populations. Seven of the closed-ended questions were multiple choice with varying response options. There was one additional closed-ended ranked question that was converted into a (weighted) Borda count. The two open-ended responses were converted to a frequency table. Using the chi-squared test individually on each question, we found that four out of 10 questions had statistically significant differences between groups (using a per-test Type I error rate of 0.05). However, after accounting for the family-wise error using the Holm-Bonferroni adjustment for all 10 questions, we found that there were no questions with a statistically significant difference in responses (using a family-wise Type I error rate of 0.05). Response bar graphs for all common closed-ended questions, along with the p-values and adjusted significance thresholds for these questions, are listed in Figures 1, 4-7, and 9-11. The two open-ended question frequency tables are shown in Tables 2 and 3. These results are discussed below.

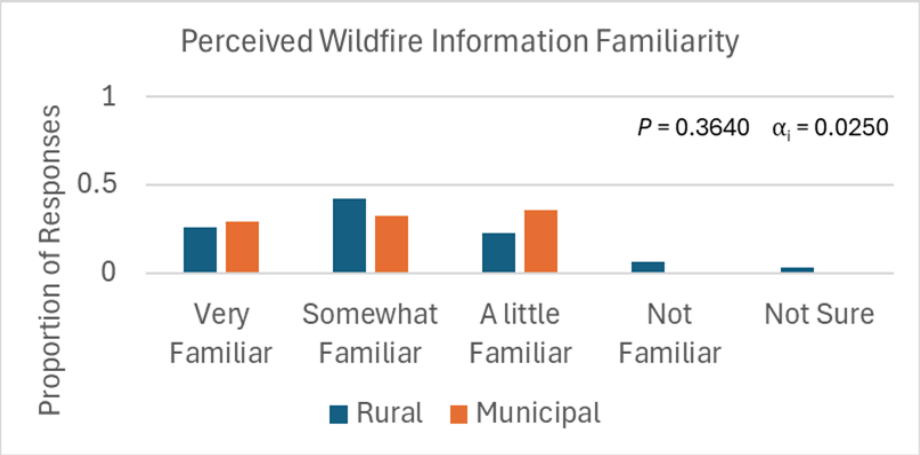
Wildfire Knowledge

For general wildfire knowledge, we asked participants to report how familiar they were with wildfire information in a multiple choice question. Figure 4 shows the perceived level of

familiarity for both rural and municipal populations, represented as the percentage of total responses for each group. Notably, over 60% of both populations felt that they were somewhat or very informed about wildfires.

Figure 4

Perceived Wildfire Information Familiarity

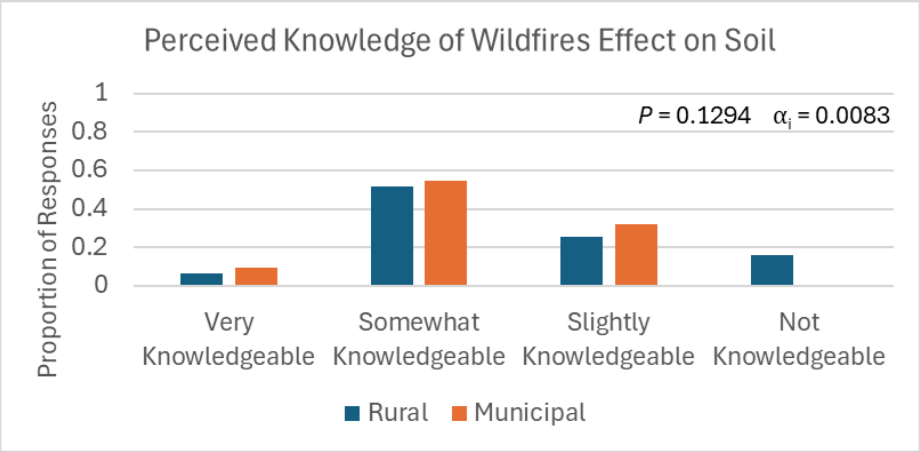


Note. This chart shows how knowledgeable participants perceive themselves to be. The X-axis is the response options for their level of familiarity with wildfire information, and the Y-axis is the proportion of total responses.

In addition to general wildfire knowledge, we were also interested in specific soil science knowledge. Figure 5 shows the level of knowledge recipients feel they have on wildfires' effects on soil. In contrast to general knowledge of wildfires, respondents both in the rural and municipal groups felt less knowledgeable about fire-effects on soil, with the fraction who felt very knowledgeable constituting less than 10%.

Figure 5

Perceived Knowledge of Wildfire Effects on Soil

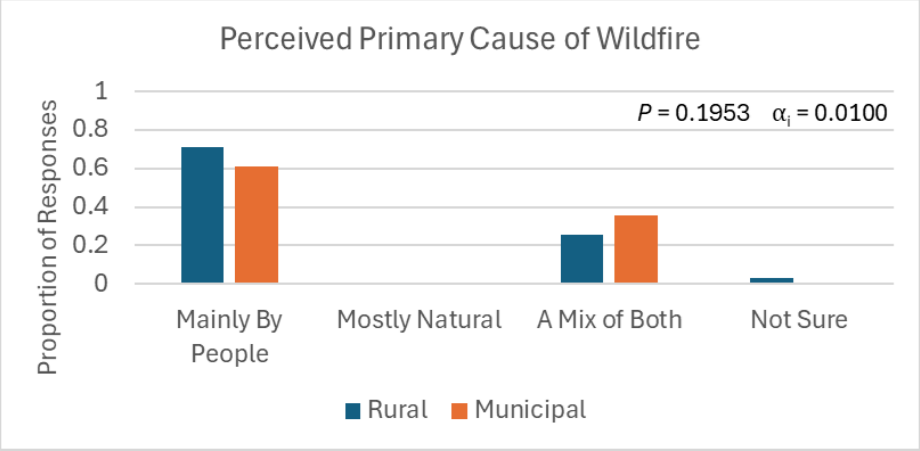


Note. This chart shows how knowledgeable participants perceive themselves on soil. The X-axis is the response options for their level of perceived knowledge, and the Y-axis is the proportion of the total responses.

Participants were asked what they attribute the primary cause of wildfires to. Figure 6 shows their responses. The majority of both populations agree that the primary cause is human-initiated fires. These reported perceived causes were compared with findings from a case study that found that humans caused 87% of wildfire ignitions (Harrison et al., 2025).

Figure 6

Perceived Primary Cause of Wildfires



Note. This chart shows the perceived primary cause of wildfire. The X-axis is the response options for causes, and the Y-axis is the proportion of the total responses.

For knowledge, both groups generally believe themselves to be at least somewhat knowledgeable about general and soil-specific effects from wildfires. Only 6% of rural participants and no municipal participants stated they were not familiar with wildfire information, and for soil-specific knowledge, just 16% of rural respondents reported not being knowledgeable. To further improve our understanding of community knowledge, the open-ended questions gave deeper insight into the effects of wildfires in general and specifically on soils.

As mentioned previously, we used grounded theory to determine categories for the general effects of wildfires and the specific effects on soil from the answers to the open-ended questions. Subsequently, frequency analysis was used to convert this qualitative data into a quantitative frequency table for both populations, where the chi-squared test was applied. Similar categories were combined so that there were common categories between the groups, such as climate impacts and environment, to improve the quality of analysis. Similar categories allowed the tables for each population to be combined into one. The final table for the general effects of wildfires can be seen in Table 2. A chi-squared test gave a p-value of 0.0333, which is greater than the adjusted significance threshold of 0.0063.

Table 2

Reported General Effects of Wildfires (Reproduced from Executive Summary)

Reported General Effects of Wildfires		
Category	**Rural Frequency**	**Municipal Frequency**
Animals	15	14
Water	12	5
Vegetation	10	15
Air quality / Environment / Climate Impacts	8	11
Biodiversity / Ecosystem	5	12
Agriculture / Livelihoods	4	0
Soil degradation	3	9

Note. This table shows the frequency of categories mentioned within the open-ended question pertaining to the general effects of wildfires for rural and municipal populations.

Similarly, for the other open-ended question on soil-specific effects of wildfires, frequency analysis was used, similar categories were combined, and the tables were grouped into

one with both populations. The combined table can be seen in Table 3. A chi-squared test gave a p-value of 0.2974, which is greater than the adjusted significance threshold of 0.0167.

Table 3

Reported Effects of Wildfires on Soil (Reproduced from Executive Summary)

Reported Effects of Wildfires on Soil		
Category	**Rural Frequency**	**Municipal Frequency**
Vegetation recovery	8	9
Soil fertility / Nutrients	7	12
Soil erosion / Physical degradation	6	13
Soil microorganisms / Biology	3	11
Water retention	3	10
Biodiversity / Ecosystem	0	8
Soil structure and chemical properties	5	8

Note. This table shows the frequency of categories mentioned within the open-ended question pertaining to the specific effects of wildfires on soil for rural and municipal populations.

When considering specific changes observed in soils post-wildfire, rural populations noted the effects on vegetation recovery, soil nutrients, and soil erosion as their main concerns. Municipal representatives, on the other hand, stressed the physical degradation of soil the most, followed by impacts on soil fertility and microorganisms.

Comparison to Existing Soil Data

Taking the observed soil effects data further, the reported effects from Table 3 were compared with quantitative data from a 2022 study on the effects of wildfire severity on the physical-chemical properties of soil in southern Ecuador, which stated that high-severity burns in 2019 were “negatively correlated with variables related to soil chemical fertility” (Carrión-Paladines et al., 2022). Consequently, it was found that people’s perceptions with respect to soil degradation align with reality.

Significant parallels can be drawn between our findings and previous studies. Agricultural producers can generally be observed to rely more heavily on qualitative indicators when making decisions based on soil. For instance, Amazonians may determine that certain black or brown soil contains earthworms based on the color (Jiménez et al., 2024). From survey responses, we likewise determined how farmers identify soil degradation based on visually

observable soil characteristics. As for communities' perceptions of soil degradation, these views typically align with the reality of the situation. Existing data states that people's perceptions in this regard agree with technical data (Jiménez et al., 2024). In addition to observed soil degradation, participants reported less confidence in soil-specific knowledge than general effects (Figures 3 and 4), but could still list many physical effects on the soil. This supports Jimenénez's findings, where agricultural producers had a strong understanding of soil conditions, but did not frame their knowledge in a technical way.

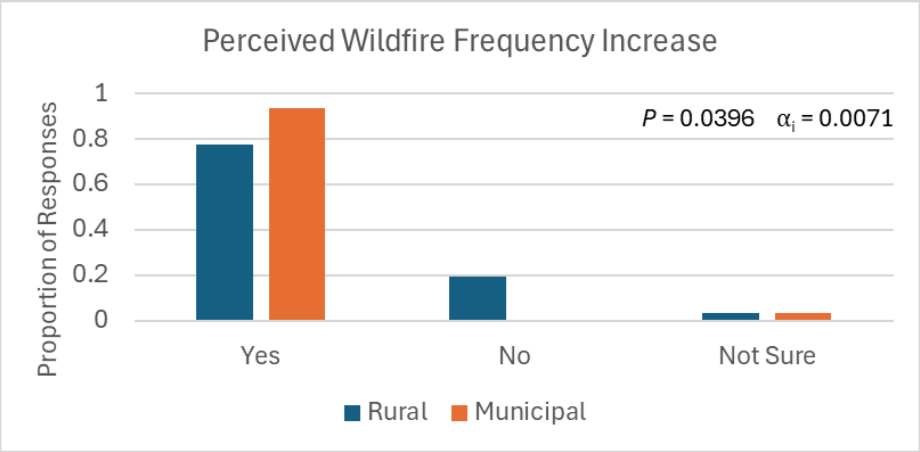
This is further corroborated by our respondents' answers congregating on concerns of soil losing its fertility, which concurs with Juan Cáceres' finding of decreased Soil Organic Carbon and pH levels in burn areas (Cáceres, 2022). Survey respondents also noted a loss of plant life and affected recovery rate following a fire. Cáceres provides a quantitative foundation for this observation by mentioning that burn sites typically experience 8.33 times less aerial plant biomass when compared with unaffected areas for a period of 18 months after a fire (Cáceres, 2022). Another shared point with Cáceres' study is what people perceive to be the principal cause of wildfires. The general consensus was that fires are primarily initiated by humans (Figure 5), which aligns with the claim that fires in El Cajas' Dos Chorreras sector are likely caused by local residents grazing horses (Cáceres, 2022).

Perceived Risk

To better understand overall perceptions of wildfires, a similar process was used to discover perceived wildfire frequency change in recent years. Figure 7 shows whether participants believe wildfires are occurring more frequently for both groups. Over 75% of both groups felt that wildfires had become more frequent in recent years.

Figure 7

Perceived Changes in Wildfire Frequency in Recent Years



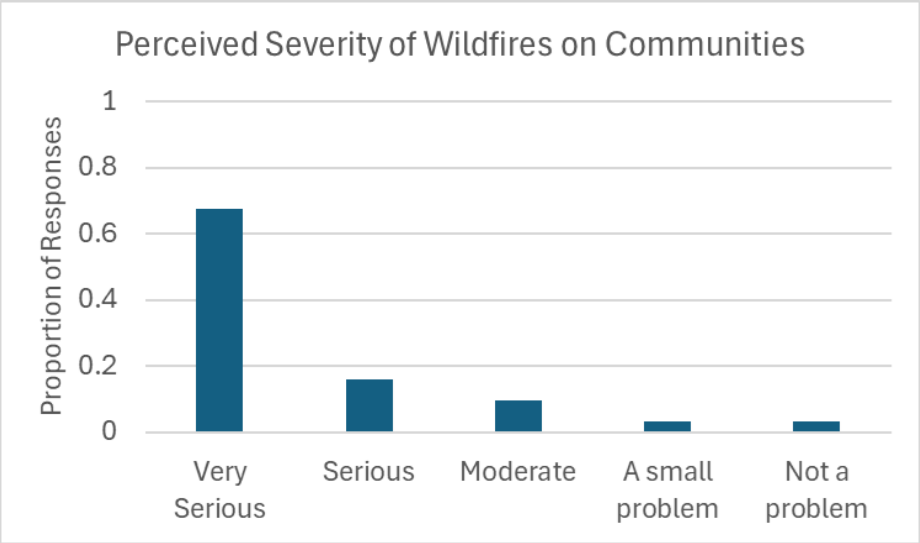
Note. This chart shows whether participants have perceived an increase in wildfire frequency. “Yes" indicates they believe there has been an increase in frequency. The X-axis is the response options for perceived increase, and the Y-axis is the proportion of the total responses.

Comparing this perception to reality, a recent study published in 2024 found that wildfire frequency over agricultural land had seen an upward trend between 2001 and 2020, totaling 440, 772 ha after the full period (Vaca et al., 2024). This change supports the idea that wildfires have become more frequent in recent years.

In addition to perceived frequency, rural community members were asked how serious they believe wildfires are for their communities. Figure 8 shows that about 84% of the individuals from rural communities perceived the risk of wildfires to be serious or very serious.

Figure 8

Perceived Severity of Wildfire's Effects on Nearby Communities



Note. This chart shows the perceived risk wildfires pose to communities. The X-axis is the response options for perceived severity level, and the Y-axis is the proportion of the total responses.

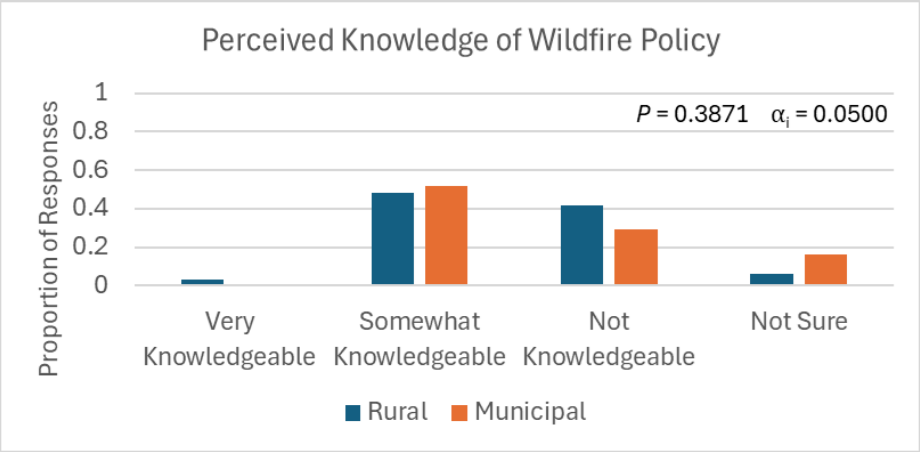
Additionally, within rural communities, they generally felt that wildfires posed a very serious risk, with 68% reporting the threat to be very serious on their communities, and only 6% considered it a small or nonexistent level of risk. This feeling of risk is backed by their knowledge on effects of wildfires in general and specifically on soil. Both groups had a strong grasp of the effects wildfires can bring.

Perceptions Surrounding Wildfire Policy

A number of our survey questions for both rural communities and municipal representatives focused on their knowledge and views on wildfire management policy. In Figure 9 below, we asked respondents to rate their knowledge of wildfire policy. Interestingly, only 3% of rural members and no one in the municipal group felt very knowledgeable about wildfire policy. This graph highlights a lack of deep knowledge and a need for more widespread information on the topic.

Figure 9

Perceived Level of Knowledge on Wildfire Policy

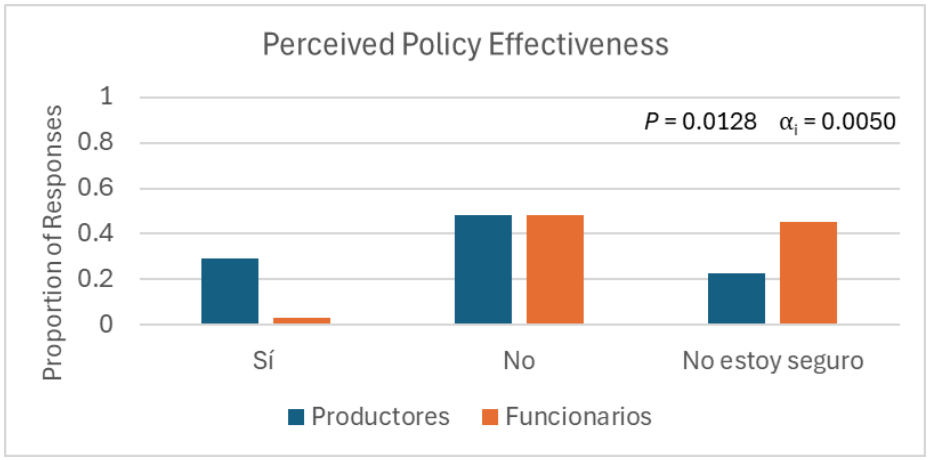


Note. This chart shows the perceived risk wildfires pose to communities. The X-axis is the response options for perceived severity level, and the Y-axis is the proportion of the total responses.

Figure 9 shows participants' beliefs in policy effectiveness. In both populations, almost 50% reported that they thought wildfire policy was not effective. While not statistically significant, a higher proportion of rural respondents answered that they believed policy was effective, while municipalities were more likely to say they were unsure. This information, combined with Figure 10, shows that many respondents have both a lack of knowledge in policy as well as an inherent distrust. This could be due to numerous factors, including issues with the current political landscape and distrust of the government as a whole.

Figure 10

Perceived Policy Effectiveness

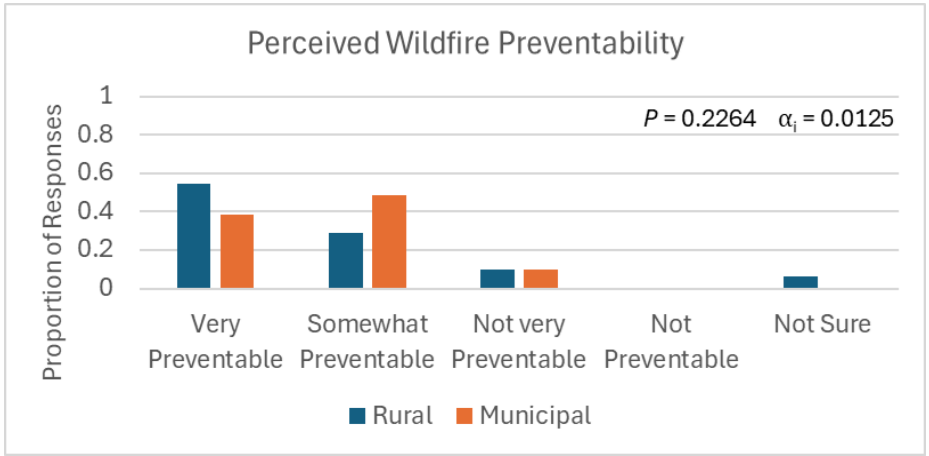


Note. This chart shows whether respondents thought policy was effective. The X-axis is the response options for if the respondent thought policy was effective, and the Y-axis is the proportion of the total responses.

To provide more context, we asked respondents how preventable they consider wildfires to be. Figure 11 shows that over 80% of both populations believe wildfires are preventable, demonstrating a belief that effective policy and practices have the capacity to improve wildfire management. The issue is not a lack of belief in a potential solution, but a lack of belief in the current solution.

Figure 11

Perceived Level of Wildfires Preventability

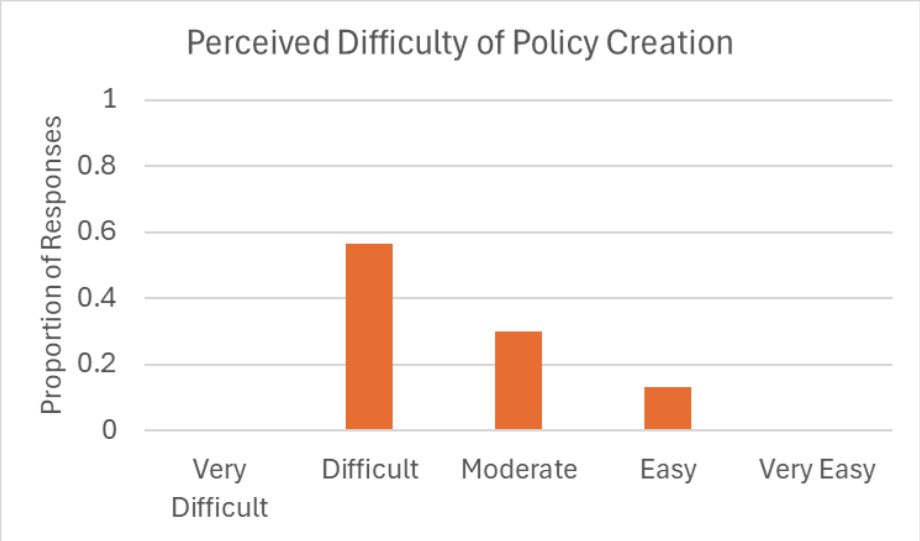


Note. This chart shows the perception of how preventable wildfires are. The X-axis is the response options for perceived preventability level, and the Y-axis is the proportion of the total responses.

Municipal workers were asked how difficult it is to create a new policy. Figure 12 shows how the municipal population regards the ease of policy creation. Most representatives, 57% of those surveyed, said policy creation is difficult, and all perceived some amount of difficulty, meaning that simplifying the process for creating policy and making it more transparent could increase support of policies.

Figure 12

Perceived Difficulty of Policy Creation

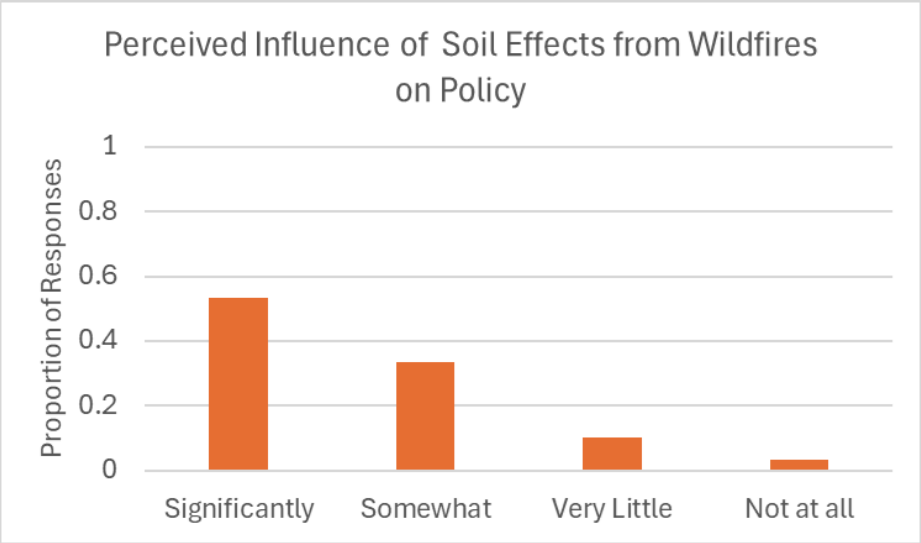


Note. This chart shows how difficult municipal representatives view creating new policies. The X-axis is the response options for how difficult policy creation is, and the Y-axis is the proportion of the total responses.

We also wanted to know how the effects of fire on soil play a role in policy creation. Figure 13 shows how municipal respondents feel the effects of wildfires on soil influence policy. This data shows that overall the effects on soil are considered important for generating policy, once again underscoring the importance of understanding wildfire effects on soil and soil research.

Figure 13

Perceived Influence of Soil Effects from Wildfires on Policy

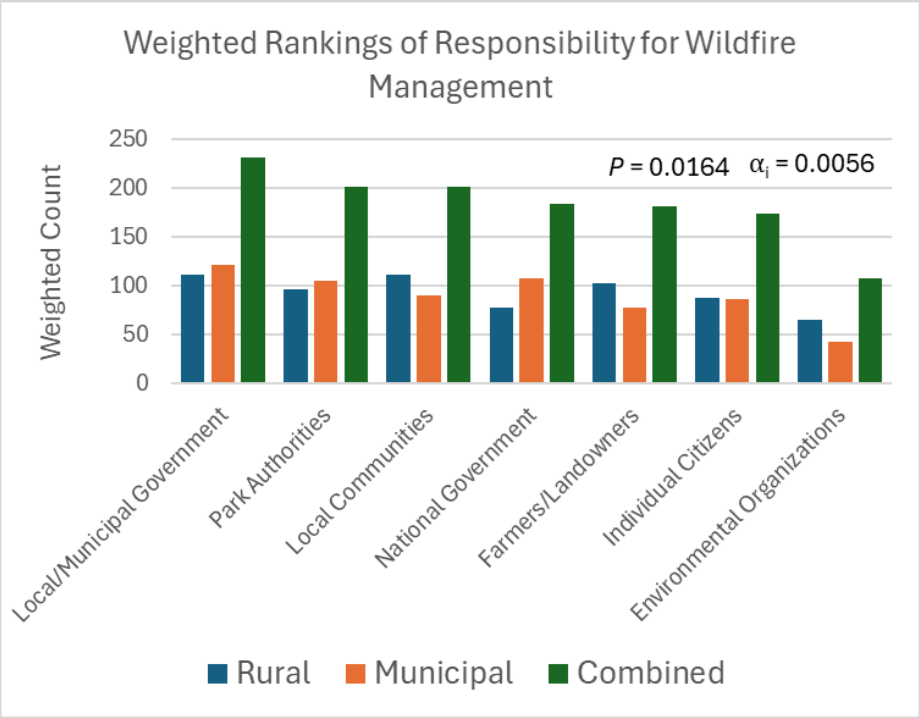


Note. This chart shows how much the effects of wildfires on soil influences the creation of policy from the perspective of municipal representatives. The X-axis is the response options for how significant a role soil plays in policy creation, and the Y-axis is the proportion of the total responses.

Individuals were also asked to rank seven options from most to least responsible for wildfire management. For analysis, a Borda count was used to produce a weighted total count, where higher ratings are associated with a higher count. Figure 1 shows the total weighted count for the rural and municipal populations, as well as the combined score. Both groups ranked local governments as the most important and non-governmental environmental organizations as the least important. The combined count on the figure reveals that while some groups may be considered more important, the responsibility of wildfire management does not fall on a single organization's shoulders. Effective management must include groups working together on education and policy. The local government plays a different role than the national government, which is again different from local communities, but all three of these groups play a crucial role in this issue.

Figure 1

Weighted Rankings of Responsibility for Wildfire Management (Reproduced from Executive Summary)



Note. This chart shows the weighted results of what rural and municipal populations believe is most responsible for managing wildfires, presented in descending combined weighted count. The X-axis is the response options for possible responsible parties, and the Y-axis is the weighted count of the survey results.

CONCLUSIONS

Our main goal for this project was to better comprehend the perceptions and experiences of our target populations in Cuenca Canton with respect to wildfires, their effects on soil, and wildfire management policy. We collected social data through the use of different surveys and traveling around Cuenca Canton to markets and governmental buildings to interview key stakeholders. Once data was collected, we cleaned it and then completed a thorough statistical analysis, including comparing and contrasting responses of rural community members and municipal representatives. In addition, we compared existing laboratory soil data from previous studies as well as previous surveying efforts to improve the robustness of our findings.

Our data suggested that gaps exist among both rural community members' and municipal representatives' knowledge on the technical impacts on soil post fire. Stakeholder groups also hold low confidence in fire management policies and their effectiveness. By contrasting our findings with previous research conducted in the same region, these studies together underscore the importance of strengthening connections between scientific research, community knowledge, and municipal decision-making. Our findings provide a baseline for future iterations and extensions of this study.

Limitations

Despite obtaining notable results from our data collection and analysis, there were limitations in our project due to our available resources and timeframe. The seven-week time frame necessitated the use of convenience sampling, leading our sample populations to not be as representative as other sampling styles, such as random or stratified random sampling. In addition, a language and cultural barrier restricted the number of respondents we could survey. This small sample size limited our statistical power. A larger sample size would have more statistical power, and it might be able to better identify or rule out statistical differences between groups.

By nature, inquiring about people's perceptions of wildfires and their effects is highly subjective, and respondents are not always completely truthful. Finally, our soil data was taken from a previous student of our sponsor from a thesis in 2022 (Cáceres, 2022). While his data proved useful, additional current soil data from El Cajas from multiple studies would improve

findings. His report also cites a lack of available information on soil data in El Cajas as a difficulty, which was also notable in our project.

Recommendations

The work, analyses, and findings resulting from this study can be used as a reference for subsequent research in this area and context. Relevant questions can be included to effectively bridge the gap between technical and social data at the time of analysis, during which subjective perceptions must be compared and contrasted with strictly objective scientific findings. In analyzing said data, evidence can be provided for the improvement of existing fire management policies. Management policies also need to be more widely understood. This can be achieved by policies being more transparent and clearly presented to the public.

The observed gaps in knowledge can serve as educational goals to be considered when creating or modifying school- or other educational programs. On the more technical side, additional soil properties can be chosen as subjects of study that more strongly support and integrate with the desired social data. Future efforts may elect to perform studies on other environmental factors similarly affected by wildfires.

Future research should aim to use more systematic sampling methods to improve representativeness and allow for stronger comparisons across stakeholder groups. Expanding the sample size and including additional rural communities would strengthen conclusions as well. Surveying other key groups, such as academics and urban residents, could provide additional important perspectives and information. Additionally, this larger sampling size would provide more statistical power and better be able to identify statistical differences between groups.

Overall, there should be more education done on wildfires and how to prevent them. One survey respondent even stated, “People should be made more aware, especially children. Ideally, the topic should be included as a subject in the school curriculum.” Further investigation into how soil science findings can be communicated to both community members and municipal representatives is also crucial. Currently, technical knowledge is mostly limited to those who have specifically studied it. This information needs to not only be readily available but also spread to improve general knowledge and understanding. To rectify this misalignment in knowledge, programs can be developed to disseminate laboratory findings in such a way that

makes sense and is informative for the general public. Visual media, such as infographics and videos, can be utilized to make known the longer-lasting and more indirect effects of fires. More hands-on programs can be designed wherein public civilians collect and measure data themselves.

BIBLIOGRAPHY

Arcentales-Ríos, R., Carrión-Méndez, A., Cipriani-Ávila, I., Acosta, S., Capparelli, M., Moulatlet, G. M., & Pinos-Vélez, V. (2022). *Assessment of metals, emerging contaminants, and physicochemical characteristics in the drinking water and wastewater of Cuenca, Ecuador*. *Journal of Trace Elements in Medicine and Biology*. https://doi.org/10.1016/j.jtemin.2022.100030

Armenteras, D., Barreto, J. S., Tabor, K., Molowny-Horas, R., & Retana, J. (2017). Changing patterns of fire occurrence in proximity to forest edges, roads and rivers between NW Amazonian countries. *Biogeosciences*, *14*(11), 2755. https://link.gale.com/apps/doc/A494561208/AONE?u=mlin_c_worpoly&sid=googleScholar&xid=73761ea8

Berrones Guapulema, G., Wilcox, B. P., Célleri Alvear, R. E., Ochoa Sánchez, A. E., & Crespo Sánchez, P. J. (2022). *Importance of fog and cloud water contributions to soil moisture in the Andean páramo*. *Hydrology, 9*(4), 54. https://doi.org/10.3390/hydrology9040054

Cáceres Lazo, J. C. (2022). *Evaluación de los efectos de un incendio en las propiedades hidrofísicas-químicas del suelo y biomasa vegetal en una zona de páramo en el área de amortiguamiento del Parque Nacional Cajas* (Undergraduate thesis, Universidad de Cuenca). Universidad de Cuenca. https://dspace.ucuenca.edu.ec/items/499beea1-1057-4682-97ea-339a09bba23a

Calvani, S., Paoloni, R., Foderi, C., Frassinelli, N., Kirschner, J. A., Menini, A., Galeotti, G., Neri, F., & Marchi, E. (2025, June 20). *Wildfire risk perception survey: Insights from local communities in Tuscany, Italy - fire ecology*. SpringerLink. https://link.springer.com/article/10.1186/s42408-025-00380-5

Carrión-Paladines, V., Fries, A., Hinojosa, M. B., Oña, A., Álvarez, L. J., Benítez, Á., Rodríguez, F. L., & García-Ruiz, R. (2023). Effects of low-severity fire on soil physico-chemical properties in an Andean Páramo in Southern Ecuador. *Fire*, 6(12), 447. https://doi.org/10.3390/fire6120447

Carrión-Paladines, V., Hinojosa, M. B., Jiménez Álvarez, L., Reyes-Bueno, F., Correa Quezada, L., & García-Ruiz, R. (2022). Effects of the Severity of Wildfires on Some Physical-Chemical Soil Properties in a Humid Montane Scrublands Ecosystem in Southern Ecuador. *Fire*, *5*(3), 66. https://doi.org/10.3390/fire5030066

Certini, G. Effects of fire on properties of forest soils: a review. *Oecologia* 143, 1–10 (2005). https://doi.org/10.1007/s00442-004-1788-8

Correa-Quezada, L., Carrión-Correa, V., López, C., Segura, D., & Carrión-Paladines, V. (2025). Towards integrated fire management: Strengthening forest fire legislation and policies in the Andean Community of Nations. *Fire*, *8*(7), 266. https://doi.org/10.3390/fire8070266

Eufemia, L., Dias Turetta, A. P., Bonatti, M., Da Ponte, E., & Sieber, S. (2022). Fires in the Amazon region: Quick policy review. *Development Policy Review*, *40*(5), e12620. https://doi.org/10.1111/dpr.12620

Harrison, S. P., Haas, O., Bartlein, P. J., Sweeney, L., & Zhang, G. (2025). Climate, vegetation, people: Disentangling the controls of fire at different timescales. *Philosophical Transactions of the Royal Society B: Biological Sciences, 380*(1924), 20230464. https://doi.org/10.1098/rstb.2023.0464

Jiménez, L., Jiménez, W., González, L., Quichimbo, P., Fierro, N., & Capa-Mora, D. (2024). Rescuing local knowledge with regards to soil management and fertility in the Amazon

Region of Ecuador. *Environmental Development, 50*, 100984. https://doi.org/10.1016/j.envdev.2024.100984

López-Gil, J. F. (2025). Addressing the impact of forest fires in Quito on youth health. *Frontiers in Public Health*, *13*, 1532865. https://doi.org/10.3389/fpubh.2025.1532865

Mistry, J., Bilbao, B. A., & Berardi, A. (2016). Community owned solutions for fire management in tropical ecosystems: Case studies from Indigenous communities of South America. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *371*(1696), 20150174. https://doi.org/10.1098/rstb.2015.0174

Moura, L. C., Pivello, V. R., Berlinck, C. N., Schmidt, I. B., & Fidelis, A. (2025). Challenges for a better fire management in South America. In A. Fidelis & V. R. Pivello (Eds.), *Fire in the South American Ecosystems* (pp. 401–417). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-89372-8_14

Nelson, A. R., Rhoades, C. C., Fegel, T. S., Roth, H. K., Caiafa, M. V., Glassman, S. I., Borch, T., & Wilkins, M. J. (2024). Wildfire impact on soil microbiome life history traits and roles in ecosystem carbon cycling. *ISME Communications, 4*(1), ycae108. https://doi.org/10.1093/ismeco/ycae108

Noble, C. D., Gilroy, J. J., & Peres, C. A. (2025). *Small forest patches and landscape-scale fragmentation exacerbate forest fire prevalence in Amazonia*. *Journal of Environmental Management, 375*, 124312. https://doi.org/10.1016/j.jenvman.2025.124312

Oliveras Menor, I., Prat-Guitart, N., Spadoni, G. L., Hsu, A., Fernandes, P. M., Puig-Gironès, R., Ascoli, D., Bilbao, B. A., Bacciu, V., Brotons, L., Carmenta, R., de-Miguel, S., Gonçalves, L. G., Humphrey, G., Ibarnegaray, V., Jones, M. W., Machado, M. S., Millán, A., de Morais Falleiro, R., … Armenteras Pascual, D. (2025). Integrated fire management

as an adaptation and mitigation strategy to altered fire regimes. *Communications Earth & Environment*, *6*(1), 202. https://doi.org/10.1038/s43247-025-02165-9

Qin, X., Wang, Y., Hou, D., & Li, Y. (2024). Assessment of high-severity post-fire soil quality and its recovery in dry/warm valley forestlands in Southwest China through selecting the minimum data set and soil quality index. *Forests*, 15(10), 1727. https://doi.org/10.3390/f15101727

Reis, S. M., Malhi, Y., Junior, B. H. M., Marimon, B. S., Zhang-Zheng, H., Araújo, I., Freitag, R., Oliveira, E. A. de, Peixoto, K. da S., Januário, L., Souza, R. D., Edim, Silva, I. L. S. da, Santos, E. B., Silva, K. P. da, Maeda, L., Gonçalves, A., … Oliveras Menor, I. (2025). Sensitivity of tropical woodland savannas to El Niño droughts. *Biogeosciences, 22*(15), 3949–3965. https://doi.org/10.5194/bg-22-3949-2025

Requia, W. J., Amini, H., Mukherjee, R., Gold, D. R., & Schwartz, J. D. (2021). Health impacts of wildfire-related air pollution in Brazil: A nationwide study of more than 2 million hospital admissions between 2008 and 2018. *Nature Communications*, *12*, 6555. https://doi.org/10.1038/s41467-021-26822-7

Santiago-Gómez, E., & Rodríguez-Rodríguez, C. (2023). Building forest fires resilience: The incorporation of local knowledge into disaster mitigation strategies. *Social Sciences, 12*(7), 420. https://doi.org/10.3390/socsci12070420

Sobreira, E., Lázaro, W. L., Vitorino, B. D., Frota, A. V. B. da, Young, C. E. F., Campos, D. V. de S., Viana, C. R. S., de Oliveira, E., López-Ramirez, L., de Souza, A. R., Silva, D. J. da, Ignotti, E., Hacon, S., Ignácio, Á. R. A., Muniz, C. C., Filho, M. dos S., & Bogoni, J. A.

(2025). Wildfires and their toll on Brazil: Who's counting the cost? *Perspectives in Ecology and Conservation*, *23*(3), 214–217. https://doi.org/10.1016/j.pecon.2025.06.003

Thompson, M. P., Calkin, D. E., Finney, M. A., Ager, A. A., & Gilbertson-Day, J. W. (2011, March 8). *Integrated National-scale assessment of wildfire risk to human and ecological values - stochastic environmental research and risk assessment*. SpringerLink. https://link.springer.com/article/10.1007/s00477-011-0461-0#Abs 1

Urrutia-Pereira, M. Rizzo, L. V., Chong-Neto, H. j., & Solé, D. (2021). Impact of exposure to smoke from biomass burning in the Amazon rain forest on human health. *Journal Brasileiro de Pneumologia, 47(5), e20210219.* https://doi.org/10.36416/1806-3756/e20210219

Vaca, C. C., Calahorrano, J., & Manzano, M. (2024). Revista Colombia Forestal. *Colombia forestal*, *27*(1), e20111. https://doi.org/10.14483/2256201X.20111

APPENDIX A: Rural Communities Survey

Percepciones de productores agropecuarios sobre los incendios forestales y el suelo

Esta encuesta forma parte de una investigación sobre los incendios forestales y sus efectos en el ecosistema, especialmente en los suelos altoandinos. El objetivo es conocer la opinión y experiencia de los productores agropecuarios sobre este tema. No hay respuestas correctas o incorrectas; nos interesa su percepción personal, basada en lo que ha vivido, observado o escuchado. Sus respuestas serán anónimas y se utilizarán únicamente con fines académicos.

Acepta participar de forma voluntaria: SÍ ___ NO ___

Fecha (día/mes/año):

Código de la encuesta:

Información personal

1. Género
    - Masculino
    - Femenino
    - Otro: _____
    - Prefiero no decir

2. ¿Cuál es su edad en años?

3. ¿Cómo se llama el sector y/o parroquia donde vive?

4. ¿Cuántos años lleva viviendo en ese sector?

5. ¿Cuál es su nivel máximo de estudios?
    - Primaria
    - Secundaria
    - Superior
    - Ninguno
    - Prefiero no decirlo

Conocimientos generales sobre incendios forestales

1. ¿Qué tan informad@ cree que está sobre los incendios forestales en general?
    - Muy familiar
    - Algo familiar
    - Poco familiar
    - No familiarizad@ en absoluto
    - No estoy segur@

2. Cree que la mayoría de los incendios forestales son:
    - Mayormente naturales
    - Causados principalmente por personas
    - Una mezcla de ambos
    - No estoy segur@

3. ¿Cree que los incendios forestales se han vuelto más frecuentes en los últimos años?
    - Sí
    - No
    - No estoy segur@

4. ¿Hasta qué punto cree que se pueden prevenir los incendios forestales?
    - Son muy prevenibles
    - Son algo prevenibles

- No son muy prevenibles
- No se pueden prevenir en absoluto
- No estoy segur@

5. En su opinión ¿Sabe qué acciones deben tomar las personas para reducir el riesgo de incendios forestales?
   - Sí—Ejemplos:

   - No
   - No estoy segur@

6. ¿Dónde ha oído o recibido información sobre incendios forestales?
   - Amigos/familiares
   - Líderes comunitarios
   - Radio
   - Televisión
   - Formación educativa
   - Funcionarios del gobierno o del parque nacional
   - No he recibido información

7. En general, ¿Qué tan graves cree que son los incendios forestales para comunidades como la suya?
   - Muy graves
   - Serios
   - Moderados
   - Son un problema pequeño
   - No son un problema
   - No estoy segur@

8. ¿Antes de esta encuesta había oído hablar de los incendios forestales en El Cajas?
   - Sí
   - No
   - No estoy segur@

9. ¿Cree que los incendios forestales en El Cajas afectan a las comunidades cercanas?
   - Sí — ¿cómo?

   - No
   - No estoy segur@

10. En su experiencia, si es que ha tenido alguna ¿En qué medida los incendios forestales le han afectado a usted o a su familia?
    - No me han afectado
    - Me han afectado levemente
    - Me han afectado de manera moderada
    - Me han afectado mucho
    - No estoy segur@

11. Si desea y solo en caso de haber sido afectad@, ¿de qué manera(s) se vio afectad@?

12. ¿Qué riesgo cree que representan los incendios forestales para usted?
    - Muy alto
    - Alto
    - Moderado
    - Bajo
    - Muy bajo
    - Ninguno
    - No estoy segur@

13. ¿Qué consecuencias en general cree usted que podrían traer los incendios forestales?

14. ¿Cuál es su conocimiento sobre los efectos que tienen los incendios forestales sobre el suelo?
    - Sin conocimiento
    - Poco conocimiento
    - Conocimiento moderado/intermedio
    - Conocimiento profundo

15. Según lo que usted ha visto, vivido o escuchado, ¿qué cambios cree usted que ocurren en el suelo después de un incendio forestal?

16. ¿Ha observado un suelo afectado por un incendio? Si es así, ¿qué ha observado?

Percepciones sobre la gestión/políticas actuales en torno a los incendios

Entendiendo que la "Política" pública es un conjunto de acciones, decisiones y estrategias planificadas por el gobierno, a menudo en colaboración con la sociedad civil, para resolver problemas públicos específicos o mejorar el bienestar social.

1. ¿Cuánto sabe sobre la política de incendios forestales?
    - Tengo mucho conocimiento
    - Conozco algunas de las políticas
    - No conozco ninguna política
    - No estoy segur@

2. ¿Cree que las políticas actuales de gestión de incendios son eficaces?
    - Sí
    - No
    - No estoy segur@

3. ¿Cómo afectan las actuales políticas de gestión de incendios en su riesgo percibido?
    - Me siento más segur@
    - Sin cambios
    - Me siento más en riesgo
    - No estoy segur@

1. ¿Quién cree que es el principal responsable de prevenir los incendios forestales en su región? Clasifique de más a menos responsable colocando la letra de cada elección en la columna de la derecha.

A. Gobierno nacional	1. ___
B. Gobierno local/municipal	2. ___

C. Autoridades del parque o ambientales	3. ___
D. Comunidades locales	4. ___
E. Agricultores/terratenientes	5. ___
F. Ciudadanos individuales	6. ___
G. ONGs u organizaciones ambientales	7. ___

APPENDIX B: Municipal Representatives Survey

Percepciones de funcionarios sobre los incendios forestales y el suelo

Esta encuesta forma parte de una investigación sobre los incendios forestales y sus efectos en el ecosistema, especialmente en los suelos altoandinos. El objetivo es conocer la opinión y experiencia de los funcionarios sobre este tema. No hay respuestas correctas o incorrectas; nos interesa su percepción personal, basada en lo que ha vivido, observado, escuchado o aprendido. Sus respuestas serán anónimas y se utilizarán únicamente con fines académicos.
Acepta participar de forma voluntaria: SÍ ___ NO ____
Fecha (día/mes/año):
Código de la encuesta:

Información personal

1. Género
    - Masculino
    - Femenino
    - Otro:_____
    - Prefiero no decir
2. ¿Cuál es su edad en años?

    _____
3. ¿Cuántos años lleva trabajando en esta área?
    - Menos de un año
    - Entre 1 y 10 años
    - Más de 10 años
4. ¿Cuál es su profesión?

    _____
5. Posee un título de educación superior de tercer nivel. En caso de ser afirmativa su respuesta ¿Cuál es su título?
    - Sí
    - No

6. Posee un título de educación superior de cuarto nivel (maestría o doctorado). En caso de ser afirmativa su respuesta ¿Cuál es su título?
    - Sí
    - No

Función laboral

1. ¿Cuál es su cargo y/o función dentro de su institución?

    _____
2. ***Entendiendo que la "Política" pública es un conjunto de acciones, decisiones y estrategias planificadas por el gobierno, a menudo en colaboración con la sociedad civil, para resolver problemas públicos específicos o mejorar el bienestar social*** ¿Qué tan implicad@ está en la elaboración de políticas?
    - Estoy directamente involucrad@
    - Tengo influencia

- No estoy involucrad@

3. ¿Ha estado o está involucrado en políticas contra incendios forestales?
    - Sí
    - Poco involucrad@
    - No

Conocimientos generales sobre incendios forestales

1. ¿Qué tan informad@ cree que está sobre los incendios forestales en general?
    - Muy familiar
    - Algo familiar
    - Poco familiar
    - No familiarizad@ en absoluto
    - No estoy segur@
2. Cree que la mayoría de los incendios son:
    - Mayormente naturales
    - Causados principalmente por personas
    - Una mezcla de ambos
    - No estoy segur@
3. ¿Cree que los incendios forestales se han vuelto más frecuentes en los últimos años?
    - Sí
    - No
    - No estoy segur@
4. ¿Hasta qué punto cree que se pueden prevenir los incendios forestales?
    - Son muy prevenibles
    - Son algo prevenibles
    - No son muy prevenibles
    - No se pueden prevenir en absoluto
    - No estoy segur@
5. ¿Cuáles son las principales consecuencias de los incendios forestales?

Hacer política

Reiterando que la "Política" pública es un conjunto de acciones, decisiones y estrategias planificadas por el gobierno, a menudo en colaboración con la sociedad civil, para resolver problemas públicos específicos o mejorar el bienestar social.

1. ¿Qué tan difícil es crear nuevas políticas?
    - muy fácil
    - fácil
    - No particularmente fácil o difícil
    - difícil

- muy difícil

2. ¿Cuánto afecta la influencia pública a su toma de decisiones sobre políticas?

3. ¿Han influido los recientes incendios en Cajas en sus prioridades a la hora de elaborar políticas?
    - Sí
    - No
4. Si es aplicable, ¿cuánto diría que ha sido influenciad@?
    - Influencia menor
    - Algo influenciado
    - Moderadamente influenciad@
    - Muy influenciad@
    - No influenciad@

Percepciones sobre la gestión/políticas actuales en torno a los incendios

1. ¿Cuánto sabe sobre la política de incendios forestales?
    - Tengo mucho conocimiento
    - Conozco algunas de las políticas
    - No conozco ninguna política
    - No estoy segur@
2. ¿Cree que las políticas actuales de gestión de incendios son eficaces?
    - Sí
    - No
    - No estoy segur@
3. ¿Quién cree que es el más responsable de prevenir los incendios forestales en su región? Clasifique de más a menos responsable colocando la letra de cada elección en la columna de la derecha

A. Gobierno nacional	1. ___
B. Gobierno local/municipal	2. ___
C. Autoridades del parque o ambientales	3. ___
D. Comunidades locales	4. ___
E. Agricultores/terratenientes	5. ___
F. Ciudadanos individuales	6. ___
G. ONGs u organizaciones ambientales	7. ___

Conocimientos sobre los impactos de los incendios forestales en los suelos

1. ¿Cuál es su conocimiento sobre los efectos que tienen los incendios forestales sobre el suelo?
    - Sin conocimiento
    - Poco conocimiento
    - Conocimiento moderado/intermedio
    - Conocimiento profundo
2. Enumere algunos efectos que tienen los incendios forestales sobre el suelo.

3. ¿Qué tan grave cree que es la degradación del suelo por incendios para Cuenca y las zonas circundantes?
    - Muy grave
    - Algo grave
    - No es muy importante
    - Nada importante
4. ¿En qué medida influyen los efectos de los incendios forestales en el suelo en la forma en que elabora o juzga las políticas?
    - Significativamente
    - Algo
    - Muy poco
    - Ninguno

Varios (pregunta abierta sobre comentarios y/o sugerencias adicionales)

1. ¿Tiene algún comentario y/o sugerencia adicional que nos quiera compartir relacionado con el tema de esta encuesta?

APPENDIX C: UConCiencia Unformatted Post

Título: Percepciones de los Incendios Forestales

Autor/a: Vicente Bastura, Roberto Carrillo Aguilar, Robert Galpern-Levin, Luis Novoa

Cuerpo o texto:
Recorrimos mercados y edificios municipales del cantón Cuenca, encuestando a un total de 61 personas. Se diseñaron encuestas con preguntas superpuestas para cada una de dos poblaciones: comunidades rurales y funcionarios municipales. Estas encuestas incluían preguntas destinadas a evaluar el riesgo percibido, la percepción de las políticas, el nivel general de conocimiento sobre incendios forestales y algunos conocimientos específicos sobre suelos. La encuesta municipal también incluía preguntas dirigidas a evaluar la comprensión de los participantes sobre la creación de políticas.

Como resultado de las encuestas y el análisis, surgieron varios hallazgos y perspectivas clave, tanto de las preguntas comunes entre la población rural y los funcionarios municipales como de las preguntas diferenciadas (Tabla 1). En cuanto a su conocimiento, ambos grupos generalmente consideran tener al menos un conocimiento moderado sobre los efectos específicos de los incendios forestales en el suelo. Solo el 16% de los encuestados rurales y ningún representante municipal reportaron no tener conocimiento. Para mejorar aún más nuestra comprensión del conocimiento de la comunidad, las preguntas abiertas brindaron una visión más profunda de los efectos de los incendios forestales en los suelos. Al considerar los cambios específicos observados en los suelos después de los incendios forestales (Tabla 2), las poblaciones rurales señalaron los efectos en la recuperación de la vegetación, los nutrientes del suelo y la erosión del suelo como sus principales preocupaciones. Los funcionarios municipales, por otro lado, enfatizaron principalmente la degradación física del suelo, seguida de los impactos en la fertilidad del suelo y los microorganismos.

Los efectos reportados en la Tabla 2 se compararon con datos cuantitativos de un estudio de 2022 sobre los efectos de la severidad de los incendios forestales en las propiedades físico-químicas del suelo en el sur de Ecuador, el cual indicó que las quemas de alta severidad en 2019 se correlacionaron negativamente con variables relacionadas con la fertilidad química del suelo (Carrión-Paladines et al., 2022). Debido a esto, encontramos que las percepciones de las personas con respecto a la degradación del suelo se alinean con la realidad. Se pueden establecer similitudes significativas entre nuestros hallazgos y estudios previos. Los productores agrícolas generalmente se basan más en indicadores cualitativos al tomar decisiones basadas en el suelo. Por ejemplo, los habitantes de la Amazonía pueden determinar que cierto suelo negro o marrón contiene lombrices de tierra según su color (Jiménez et al., 2024). A partir de las respuestas de la

encuesta, también encontramos que los agricultores identifican la degradación del suelo basándose en características del suelo visualmente observables.

Esto se corrobora aún más por las respuestas de nuestros encuestados, que se concentran en la preocupación por la pérdida de fertilidad del suelo, lo que coincide con el hallazgo de una disminución de los niveles de carbono orgánico del suelo y pH en las zonas quemadas (Cáceres, 2022). Los encuestados también observaron una pérdida de vegetación y una tasa de recuperación afectada tras un incendio. Cáceres proporciona una base cuantitativa para esta observación al mencionar que las zonas quemadas suelen experimentar 8,33 veces menos biomasa aérea vegetal en comparación con las zonas no afectadas durante un período de 18 meses después de un incendio (Cáceres, 2022).

Al considerar el riesgo percibido, los encuestados afirmaron que la frecuencia de incendios forestales había aumentado en los últimos años, con un 75% de coincidencia entre ambos grupos. Un estudio publicado en 2024 reveló que la frecuencia de incendios forestales en tierras agrícolas había seguido una tendencia al alza entre 2001 y 2020, totalizando 440.772 ha durante todo el período (Vaca et al., 2024), lo que respalda esta idea. De los encuestados, el 84% consideró que el riesgo de incendios forestales era grave o muy grave, y solo el 6% respondió que el riesgo era bajo o inexistente.

En cuanto a las políticas, los encuestados mostraron un desconocimiento profundo de la gestión de incendios forestales, así como una desconfianza inherente en la eficacia de las políticas contra incendios forestales. Esto podría deberse a numerosos factores, como problemas con el panorama político actual y la desconfianza en el gobierno en general. Sin embargo, los encuestados también afirmaron creer que los incendios forestales son prevenibles, destacando que el problema no reside en la falta de confianza en una posible solución, sino en la solución actual. Facilitar y hacer más transparente el proceso de elaboración de políticas, así como aumentar la comprensión generalizada de dichas políticas, aumentaría el apoyo a las políticas de gestión de incendios forestales.

Nuestros datos sugieren que existen lagunas en el conocimiento, tanto de los miembros de las comunidades rurales como de los funcionarios municipales sobre los impactos técnicos en el suelo tras un incendio. Los grupos de interés también tienen poca confianza en las políticas de gestión de incendios y en su eficacia. Al contrastar nuestros hallazgos con investigaciones previas realizadas en la misma región, estos estudios resaltan la importancia de fortalecer los vínculos entre la investigación científica, el conocimiento comunitario y la toma de decisiones municipal. Nuestros hallazgos proporcionan una base para futuras iteraciones y ampliaciones de este estudio.

El texto debe incluir dos o tres hipervínculos. (enlaces que nos lleven a otras publicaciones relacionadas, no necesariamente del autor/es)

- Rescuing local knowledge with regards to soil management and fertility in the Amazon Region of Ecuador - ScienceDirect
- https://dspace.ucuenca.edu.ec/items/499beea1-1057-4682-97ea-339a09bba23a
- En el futuro se publicará en Digital WPI nuestro reporte completo del proyecto con el siguiente título: Understanding Perceptions of Stakeholders in Cuenca Cantón Surrounding Wildfires

Sumilla (párrafo inicial):
Se buscaba comparar percepciones comunitarias con el análisis técnico de los efectos de los incendios forestales. Se recolectaron datos en torno a las perspectivas tanto de poblaciones rurales como de funcionarios municipales por medio de encuestas para luego analizarlos de forma conjunta. Al tener los datos analizados, se compararon e integraron con análisis de suelos previos.

Tres puntos principales (90 caracteres):
- Incendios y efectos sobre suelos: Una mayor educación sobre los incendios forestales y sus efectos sobre el suelo mejoraría la comprensión comunitaria.
- Gestión de incendios forestales: Las políticas de gestión de incendios forestales deben ser más transparentes y difundirse ampliamente.
- Creación de políticas: Las políticas deben tener en cuenta las experiencias vividas por los miembros de la comunidad y los datos técnicos del suelo.

Párrafo destacado: Los incendios forestales amenazan el Parque Nacional El Cajas, que es vital para regular la humedad global y proporcionar agua potable a Cuenca. Nuestro objetivo era comprender las percepciones y el conocimiento de la comunidad en torno a estos incendios forestales y sus efectos en el suelo.

Sobre el autor/a: Somos estudiantes del Worcester Polytechnic Institute en Estados Unidos y trabajamos en este proyecto junto con el Dr. Pablo Geovanny Quichimbo Miguitama de la facultad de agronomía.

DOI de artículos del autor/es relacionados con la temática (obligatorio):

- Rescuing local knowledge with regards to soil management and fertility in the Amazon Region of Ecuador: https://doi.org/10.1016/j.envdev.2024.100984
- Climate, vegetation, people: Disentangling the controls of fire at different timescales: https://doi.org/10.1098/rstb.2023.0464

- Effects of the Severity of Wildfires on Some Physical-Chemical Soil Properties in a Humid Montane Scrublands Ecosystem in Southern Ecuador: https://doi.org/10.3390/fire5030066
- Revista Colombia Forestal: https://doi.org/10.14483/2256201X.20111
- Evaluación de los efectos de un incendio en las propiedades hidrofísico-químicas del suelo y biomasa vegetal en una zona de páramo en el área de amortiguamiento del Parque Nacional Cajas: N/A

Metadatos y etiquetas: Suelos, recursos hídricos, percepción comunitaria, gestión de incendios



Figura 1. Ubicaciones en las que se encuestó. Los logos verdes representan mercados y los anaranjados oficinas municipales.

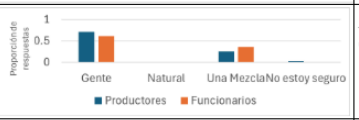
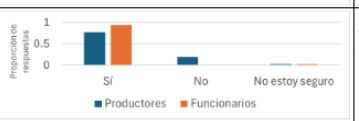
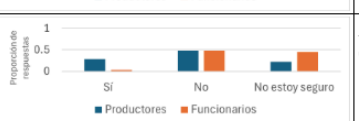
Pregunta	Gráfico	Valor P & Umbral
¿Cuánto sabes sobre incendios forestales?		$P = 0.3640$ $\alpha_i = 0.0250$
¿Cuál es la causa principal de los incendios forestales?		$P = 0.1953$ $\alpha_i = 0.0100$
¿Se han vuelto más frecuentes los incendios forestales?		$P = 0.0396$ $\alpha_i = 0.0071$
¿Qué grado de prevención pueden tener los incendios forestales?		$P = 0.2264$ $\alpha_i = 0.0125$
¿Qué nivel de conocimiento tienes sobre políticas de incendios forestales?		$P = 0.3871$ $\alpha_i = 0.0500$
¿Es eficaz la política de incendios forestales?		$P = 0.0128$ $\alpha_i = 0.0050$
¿Qué nivel de conocimiento tienes sobre el efecto de los incendios forestales en el suelo?		$P = 0.1294$ $\alpha_i = 0.0083$

Tabla 1. Las siete preguntas de opción múltiple y los resultados del análisis chi-cuadrado con el ajuste de Holm-Bonferroni.

Efectos reportados de los incendios forestales en el suelo		
Categoría	**Frecuencia Rural**	**Frecuencia Municipal**
Vegetación	8	9
Fertilidad del suelo / Nutrientes	7	12
Erosión del suelo / Degradación física	6	13
Microorganismos del suelo / Biología	3	11
Retención de agua	3	10
Biodiversidad / Ecosistema	0	8

Tabla 2. La frecuencia de categorías mencionadas en la pregunta abierta sobre los efectos generales de los incendios forestales.

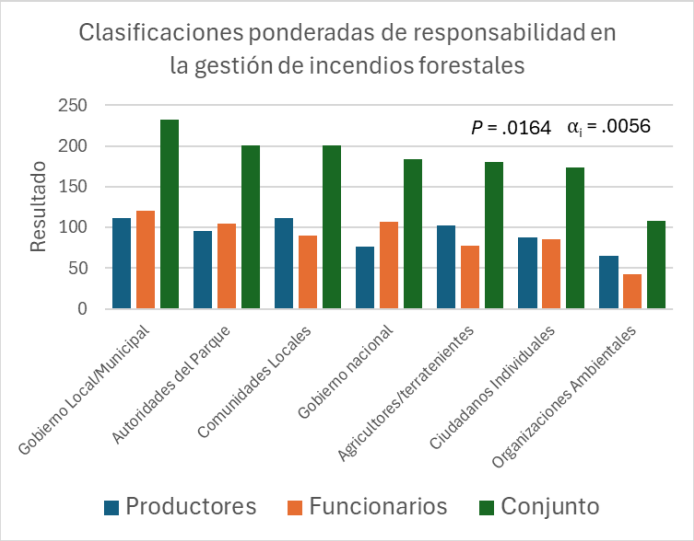


Figura 2. Los resultados ponderados de quién se considera más responsable de la gestión de incendios forestales, presentados en puntuaciones combinadas descendentes.

Proyecto al que pertenece o está relacionado el texto: N/A
Convocatoria al que pertenece el proyecto: Instituto Politécnico de Worcester & Universidad de Cuenca